

Assessing the Effectiveness of Topological Metrics for Predicting Retail Density and Categorising UK Cities

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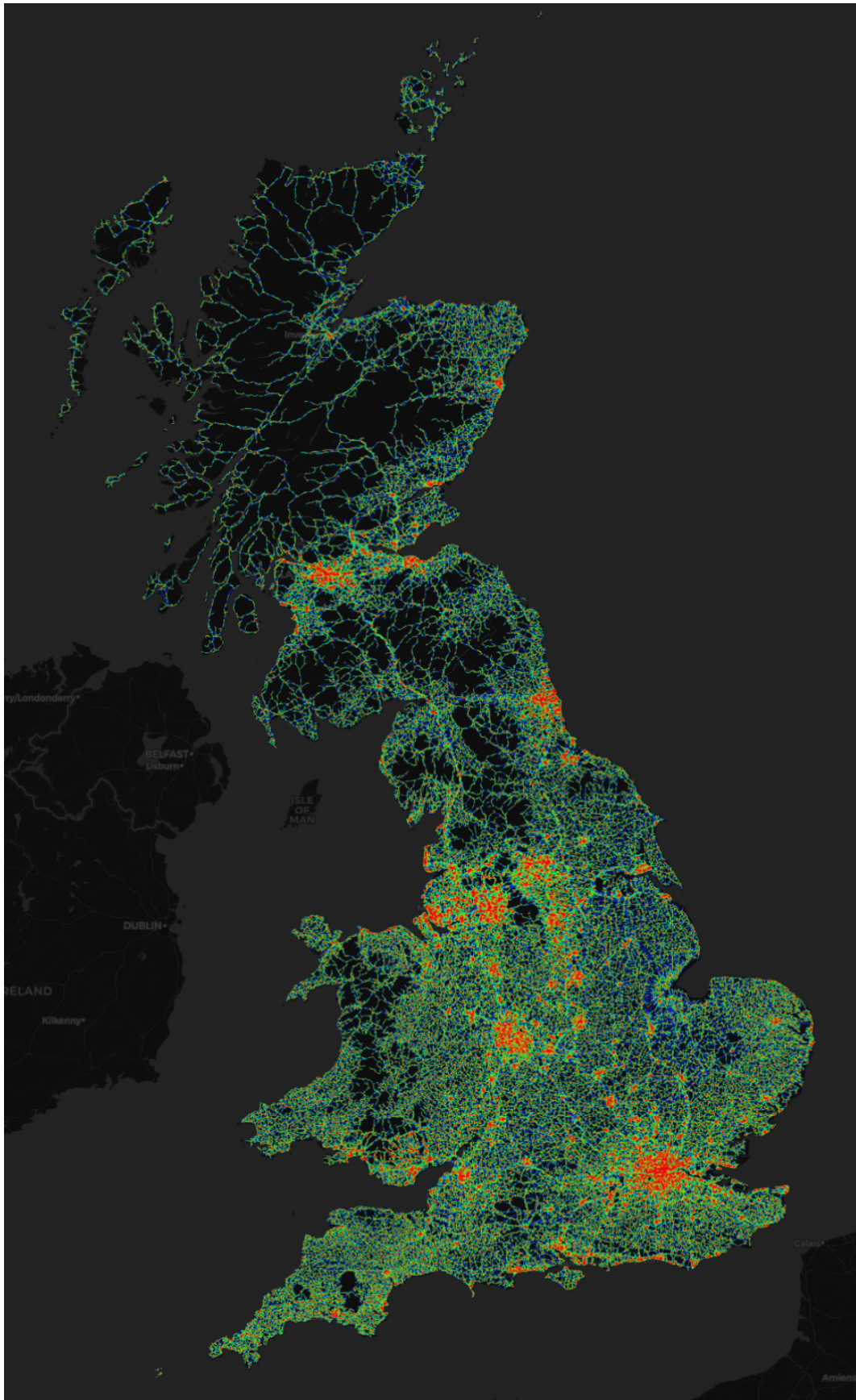
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Space Syntax's *OpenMapping*. Visualisation of Centrality Values Measured at 2km. (2018)

Abstract

This dissertation investigates the relationship between urban street network topology and retail density in UK cities. This study intends to advance our understanding of how urban form influences retail success. This research develops a typology of UK cities based on their morphological characteristics and secondly evaluates the predictive power of centrality measures in determining retail density. The research reveals that while centrality metrics are useful, they are not universally applicable predictors of retail density due to the varying urban contexts across different city categories. By integrating additional factors such as spatial autocorrelation and street network entropy, this study provides a comprehensive framework that captures the complexity of urban retail environments. The findings underscore the importance of considering a range of topological metrics to better understand and plan for the spatial dynamics of retail distribution. This study is both ambitious and one of the first of its kind in its comprehensive evaluation of urban street networks and retail densities across an entire country (UK). By leveraging modern programming libraries, this research conducts large-scale, detailed analysis that was previously unfeasible.

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The Large Language Model, ChatGPT 4, developed by OpenAI (<https://chatgpt.com/>), has been leveraged to develop and interpret portions of code used for this study. Any code developed by LLMs was rigorously sense-checked and reviewed by the author.

Declaration

I, Nikhil Desai, hereby declare that this dissertation is all my own original work and that all sources have been acknowledged. It is 11 449 words in length.

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Acronyms

ANoVA Analysis of Variance

CC Correlation Coefficient

CPU Central Processing Unit

CPT Central Place Theory

GIS Geographic Information System

Gt Graph_tool

KDC Kernel Density Correlation

KDE Kernel Density Estimate

LLM Large Language Model

MCA Multiple Centrality Assessment

MSE Mean Squared Error

OLS Ordinary Least Squares

OS Ordnance Survey

OSM Open Street Map

OSMnx Open Street Map NetworkX. Python Programming Library

PCA Principal Component Analysis

POI Points of Interest

QGIS Quantum Geographic Information System

SLAM Store Location Assessment Model

UK United Kingdom

1 Introduction

The spatial distribution of retail activities within urban environments is a key area of study in urban planning and economic geography. Understanding the factors that influence where retail clusters emerge and thrive is essential for developing effective urban strategies. One of the most critical, yet often overlooked, aspects of this analysis is the role of street network topology—specifically, how the structural characteristics of urban street networks influence retail density (Masucci et al. 2013).

Street network topology refers to the arrangement and connectivity of streets within a city, encompassing various metrics such as centrality, connectivity and complexity. These topological characteristics are increasingly recognised as powerful determinants of urban economic activity, particularly in relation to the distribution of retail spaces. Centrality measures, for instance, which assess the importance of particular streets within the overall network, have been widely used to predict areas of high retail density Porta et al. (2009), Antunes et al. (2023). The assumption is that well-connected streets with high centrality attract more foot traffic, making them prime locations for retail establishments. In 1 we see Porta’s (2009) comparison between retail density and centrality of Bologna. $1b$ represents centrality and $1d$ represents retail density.

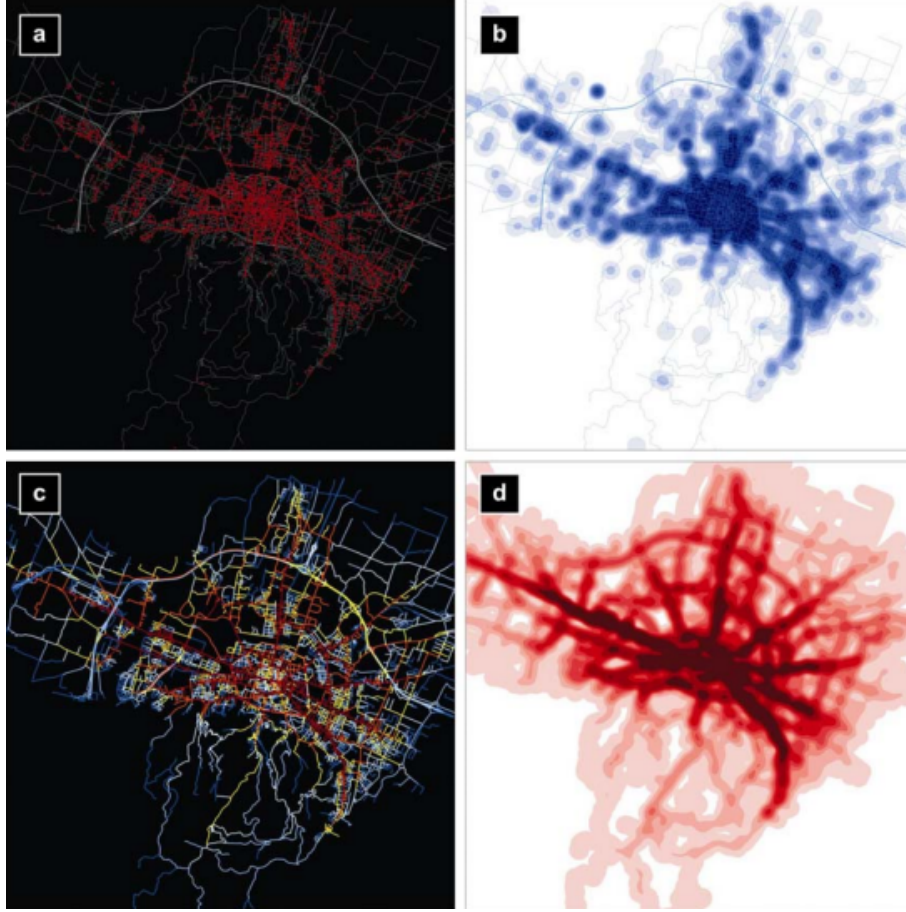


Figure 1: Density of activity and street Centrality: (a) Location of commercial and service activities (red dots); (b) Kernel Density Estimation (KDE) ($h = 300$ m) of commercial and service activities; (c) Street global betweenness (C_B^{glob}) (blue for lower values and red for higher); (d) KDE ($h = 300$ m) of C_B^{glob} . (Porta et al. 2009)

However, the relationship between street network topology and retail density is not straightforward. Recent studies have highlighted that this relationship can vary significantly depending on the specific urban context (Cui & Han 2015, Han et al. 2019a). Factors such as spatial clustering, local land use policies, historical development patterns and socio-economic conditions can all influence the effectiveness of topological metrics in predicting retail density. This complexity suggests that while topological metrics are important, they interact with a range of other spatial and non-spatial factors that collectively shape retail distribution. The results from a similar study in Zhengzhou illustrate this, showing that retail density tends to cluster in specific areas rather than align directly with high betweenness centrality (Han et al. 2019a).

Another example of this in London is Covent Garden, one of the most popular and densely populated shopping areas in London (Research 2017, Springham n.d.). In Figure 2, the perimeter of Covent Garden is outlined in below in purple, with betweenness values $C_B(v)$ indicated by the street colour—thicker and warmer tones represent higher betweenness and red dots mark retail locations. The streets within Covent Garden and nearby Soho exhibit relatively low betweenness values but extremely high retail density.

Conversely, Euston Road, a major artery in central London (seen near the top of Figure 2), sees rather low retail density but extremely high betweenness centrality.

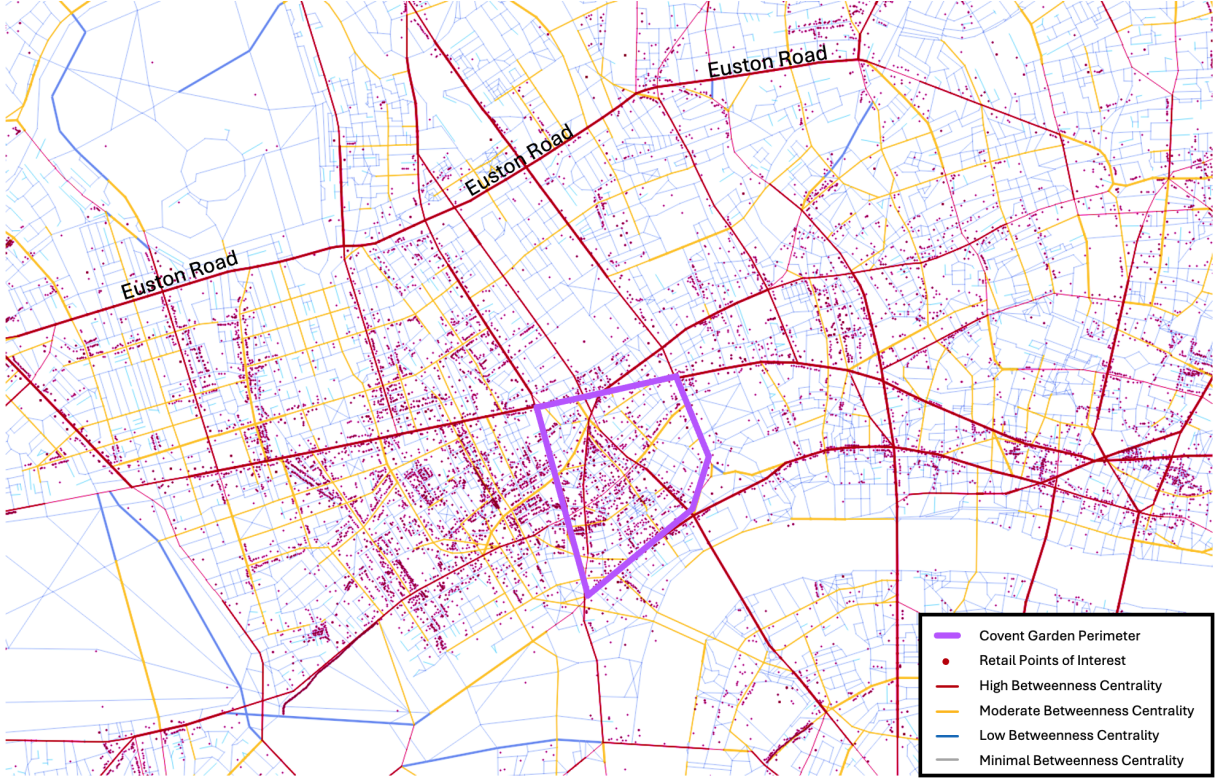


Figure 2: A map of Central London visualising normalised betweenness centrality values and retail points of interest, with the perimeter of Covent Garden highlighted in purple. Red Streets: $C_B(v) > 1.25$: Nodes with the highest betweenness centrality, critical for overall connectivity. Yellow Streets: $0.7 < C_B(v) \leq 1.25$: Nodes with moderate to moderately high betweenness centrality, important for linking local and central areas. Blue Streets: $0 < C_B(v) \leq 0.7$: Nodes with low betweenness centrality, typically peripheral or isolated. Grey Streets: $C_B(v) \approx 0$: Nodes with negligible betweenness centrality, minimally contributing to the network’s connectivity.

With this in mind, this dissertation aims to explore and clarify the relationship between street network topology and retail density within UK cities. By examining how different topological metrics—particularly betweenness centrality—correlate with retail density, this research seeks to identify the extent to which these metrics can serve as reliable predictors of economic activity. Additionally, the study will investigate how other factors, such as spatial autocorrelation and street network entropy, may further influence retail density, offering a more nuanced understanding of the spatial dynamics at play.

In addition to exploring the relationship between street network topology and retail density, this dissertation also undertakes the task of categorising UK cities based on their topological metrics rather than traditional factors like raw city size or socio-economic characteristics. This approach aims to provide a more nuanced understanding of urban form by grouping cities according to the structural properties of their street networks, such as connectivity and centrality. By focusing on topological metrics, this categorisation reveals patterns and similarities between cities that might be overlooked when only considering size or economic factors. The value of this secondary objective lies in its potential to inform

urban planning and policy-making. Because topological metrics will be calculated at the scale of the city, categorising cities based on these metrics allows for a more precise assessment of the relationship between topological characteristics and retail density. By grouping cities according to their structural properties, such as connectivity and centrality, rather than traditional factors like size or socio-economic status, this approach provides a clearer lens through which to examine how urban form influences retail distribution.

This categorisation enables a more focused analysis, revealing relationships that might be obscured when cities are treated as homogenous entities. Consequently, this secondary objective not only enhances the understanding of urban dynamics but also offers practical insights for urban planning, by identifying specific city types where topological metrics more directly impact retail density.

This approach not only deepens our understanding of urban structure and its impact on retail distribution but also sets the stage for a focused examination of these dynamics within the unique geographical framework of the United Kingdom.

1.1 Britain as a Case Study

The United Kingdom is an excellent case study for examining the relationship between urban street network topology and retail distribution due to its diverse urban forms and historical development patterns (Mohajeri, French & Gudmundsson 2013). Cities of the United Kingdom, such as London, offer a rich historical context with well-documented street network evolution, providing a unique longitudinal view of urban growth and retail dynamics (Masucci et al. 2013, Ye et al. 2020). For example, the path dependency and evolution of London’s street network showcases how historical patterns have shaped the current urban structure and connectivity. This variety allows for a detailed analysis of how different street network configurations influence retail patterns within a single national context. For instance, Sheffield’s landscape-driven street geometry significantly impacts the spatial distribution of retail stores, while older cities with organically developed networks show a more integrated retail activity compared to newer, planned cities where retail tends to be concentrated in accessible but less integrated locations (Mohajeri & Gudmundsson 2014, Omer & Goldblatt 2016).

1.2 Calculating Topological Metrics Using *Graph_tool*

Until now, studying the correlation between street network topology and retail density has primarily been conducted singularly in specific cities, thus, there has never been a country-wide study on this relationship, making this paper the first of its kind. This is a result of historical computational limitations in spatial data science and the infancy of libraries that are able to process network data. *NetworkX*, the most popular and user-friendly python library for processing graph data, is a pure Python library, not optimised for performance or speed. This results in long processing times and computationally intensive calculations which require significant CPU capacity (Azizul Hakim 2023). *Graph_tool*, a new library,

publicly released in May 2022, was developed by Tiago P. Peixoto. *Graph_tool* is significantly faster due to it being implemented in C++ using the Boost Graph Library (Peixoto 2014). This allows it to leverage the performance advantages of C++, such as faster execution and better memory efficiency (Azizul Hakim 2023). Additionally, graph-tool supports parallel computation, making it far faster for large-scale graph processing (Lin 2019, Azizul Hakim 2023). The process for converting our data into a **Graph_tool graph** (*gt graph*). is included in Appendix B.

This study leverages graph.tool to quickly assess betweenness at a country-level, making this a unique and ambitious study of centrality and retail density.

2 Research Overview

2.1 Purpose of Research

This research is motivated by an interest in exploring the structural elements of cities that influence retail distribution. Through better understanding the interplay between street networks and retail location patterns, the study seeks to contribute to the field of urban planning by offering insights that could inform more effective urban design and economic development strategies.

2.2 Scope

The scope of this dissertation encompasses an analysis of the morphological characteristics of UK cities and their impact on retail density. The research is structured in two primary parts:

1. **Clustering of UK Cities Based on Morphological Measures:** This component of the study involves categorising UK cities based on their street network topologies, independent of size. The analysis employs a range of topological metrics, including clustering coefficients and street orientation, to identify distinct categories of cities. This process aims to uncover underlying patterns and commonalities that may not be apparent when cities are considered individually.
2. **The Role of Morphological Metrics in Predicting Retail Density:** This section aims to reveal how the connectivity and positioning of streets impact the concentration of retail activities, shedding light on the spatial dynamics of urban commerce.

2.3 Objectives and Research Questions

With this in mind, this study endeavours to answer three research questions

1. **How Effective are Topological Metrics, Particularly Centrality, in Predicting Retail Density?** The primary objective of this study is to evaluate whether betweenness centrality is an effective predictor of retail density across different city types.

2. **Can UK cities be effectively categorised based on street network topologies?** The secondary objective is to develop a classification system for UK cities that relies on their morphological characteristics.
3. **Do Entropy and Spatial Autocorrelation Influence the Accuracy of Retail Density Predictions?** This research question seeks to understand whether the complexity and spatial relationships within urban street networks affect the ability to accurately predict where retail stores concentrate.

2.4 Outline

The structure of the dissertation is as follows:

1. **Introduction and Background:** This chapter introduces the research topic, outlining the significance of understanding the relationship between urban street network topology and retail distribution.
2. **Literature Review:** These chapters examine existing studies on urban morphology, centrality measures and retail density and the relevance of the UK as a case study. It highlights the theoretical foundations of the research and identifies gaps in the current knowledge that this dissertation aims to address.
3. **Methodology:** This chapter details the methods and analytical tools used in the research, including data collection, clustering techniques and statistical analyses. It explains the rationale behind the chosen methods and discusses how they were applied to achieve the research objectives.
4. **Results and Analysis:** This chapter presents and analyses the findings of the research. It includes a detailed discussion of the clustering of UK cities, the relationship between centrality measures and retail density and the identification of additional factors influencing retail distribution.
5. **Discussion and Conclusion:** This chapter interprets the results in the context of the broader literature, discussing the implications of the findings for urban planning and policy-making. It also addresses the limitations of the study and suggests areas for future research.

3 Background: Historical Methods of Assessing Retail Location

The assessment of retail location in the UK has evolved significantly over time, paralleling broader developments in urban planning aimed at fostering economic activity. The interplay between these two areas reflects changing priorities and advancements in both retail strategies and urban development practices. This evolution is summarised in the following section.

3.1 Historical Methods

In the 1970s, retailers primarily relied on subjective checklists and analogues to assess potential store locations, focusing on practical, observable factors such as pedestrian traffic, store visibility, and proximity to competitors. Urban planners during this period similarly employed pragmatic approaches, often prioritising physical infrastructure and accessibility in efforts to stimulate economic growth. However, both fields faced limitations in scope; while retailers lacked rigorous analytical tools, urban planners often operated within frameworks that did not fully account for the complexities of socio-economic interactions and spatial dynamics.

As the retail landscape and urban environments grew more intricate with increased competition and consumer mobility, the need for more advanced analytical tools became apparent. Spatial interaction models, such as those based on Reilly's Law of Retail Gravitation (Reilly 1931), were developed to predict retail performance by examining the interactions between retail locations and their surrounding catchment areas. Urban planners began to adopt similar models to better understand and influence the distribution of economic activities within cities, though the complexity of these models sometimes limited their practical application.

Central Place Theory has been foundational in both urban planning and retail location analysis, particularly within the UK. Developed by Christaller (1933), this theory posits that urban centres are organised hierarchically, with larger "central places" providing a broad range of goods and services, while smaller, peripheral locations cater to more localised, everyday needs. Urban planners have used this framework to guide the development of urban centres, ensuring that economic activities are strategically distributed to support both local and regional economies. Figure 3 visualises Christaller's Central Place Theory as applied in Germany, illustrating this hierarchical distribution.

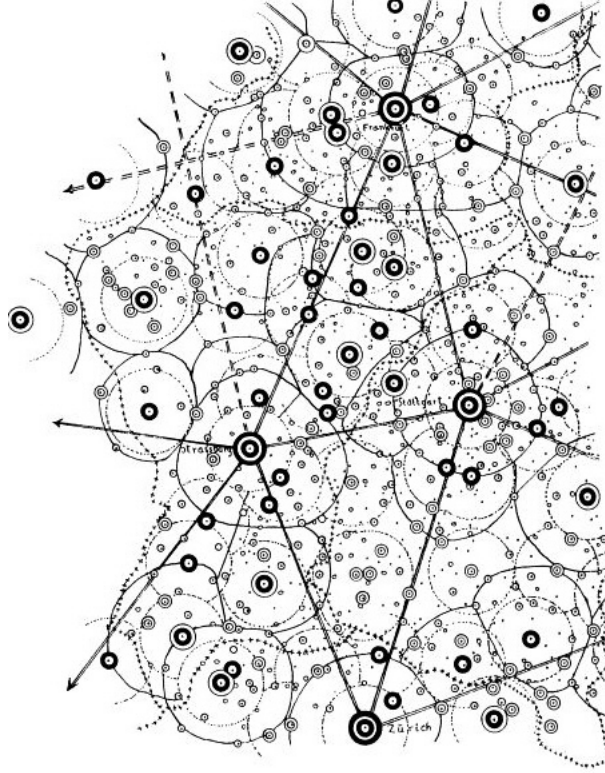


Figure 3: Illustration of Christaller's Central Place Theory (Christaller & Baskin 1933)

This framework has been instrumental in explaining the distribution and interaction of retail centres, suggesting that consumers travel to central places for specialised goods and services that are not available in smaller, nearby locations. Dennis et al. (2002) expanded on these principles by applying them to the UK context, using metrics such as retail attractiveness and hinterland boundaries to delineate the retail hierarchy. Their studies demonstrated the utility of Central Place Theory in understanding retail distribution patterns across varying urban and rural landscapes. However, they also underscored the limitations of applying such static models to dynamic and rapidly evolving urban environments, where factors like changing consumer behaviour and urban sprawl continuously reshape retail landscapes. These studies highlight the need for more flexible and adaptive models that can account for the complexities of modern retail ecosystems (Dennis et al. 2002, Huff 1964). Figure 4 illustrates the perceived catchment area around 2 examples of *central places*.

3.2 Contemporary Methods

The advent of Geographical Information Systems (GIS) revolutionised both retail location assessment and urban planning by enabling more dynamic and spatially accurate analyses. For retailers, GIS provided tools to integrate various data layers such as demographics, transportation networks, and land use patterns, thereby enhancing decision-making processes. Urban planners similarly benefited, using GIS to model urban growth, predict economic hotspots, and optimise land use planning to encourage balanced

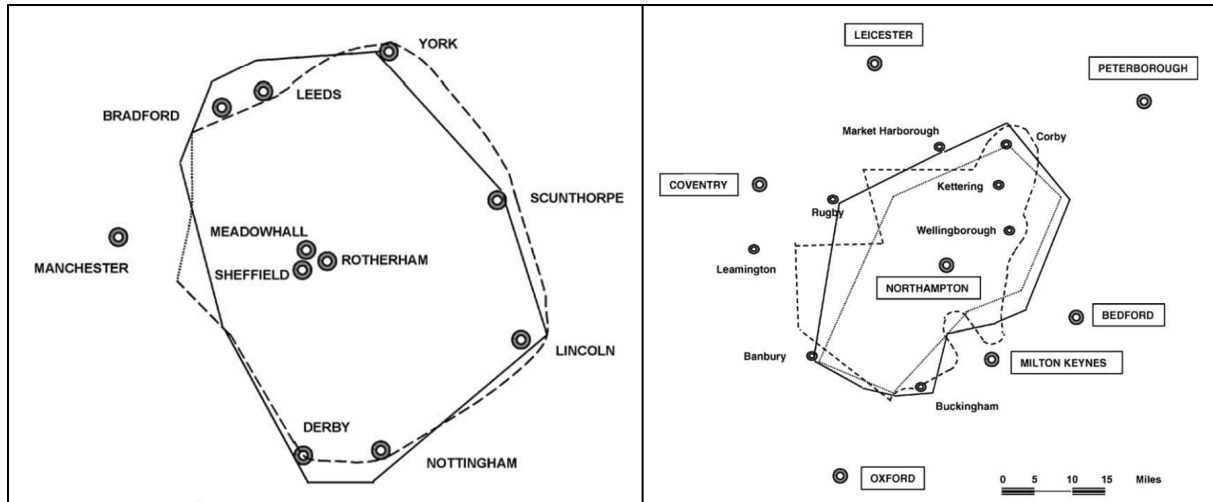


Figure 4: Charles Dennis' application of Central Place Theory to Northhampton and Sheffield, UK (2002)

economic development (Simkin 1990).

Building on these foundations, the application of network sciences and space syntax has marked a significant advancement in urban studies. These methodologies allow for a detailed analysis of urban networks, providing insights into how street connectivity, pedestrian flow, and spatial hierarchies influence retail distribution and economic activity. By leveraging advanced tools such as deep learning and space syntax, both retailers and urban planners can now model and predict urban dynamics with greater accuracy, offering a more comprehensive understanding of how urban environments support retail activity and broader economic functions. This approach moves beyond historical methods, helping to optimise urban planning and retail strategies to better align with the complex systems that underpin urban economic and social functions Ye et al. (2020), Porta et al. (2009), Stamos (2023).

4 Literature Review

4.1 Categorising UK Cities Based on Topological Characteristics

Categorising cities based on their morphological characteristics, particularly street network topologies, is useful for developing a nuanced framework for urban analysis. Traditional methods often rely on metrics like population size or economic output, which can overlook the complexities of urban form and spatial structure. However, recent advancements in urban morphology have introduced more sophisticated approaches that consider the physical layout and configuration of cities, offering a more detailed and context-sensitive categorisation.

A significant contribution to this field was made by Yamaoka et al. (2021). who used local betweenness centrality to analyse the morphological structure of 30 European cities. This study identified common structural patterns in street networks, indicating that cities with similar centrality patterns share under-

lying urban forms. This insight is particularly relevant to the categorisation of UK cities, where applying centrality measures can reveal morphological clusters that traditional methods might overlook. However, unlike Yamaoka et al.'s study, which focused on a limited number of European cities, this study systematically examines the entire country, providing a comprehensive analysis at a national scale that accounts for the diverse hierarchical urban forms across a particular country.

Advancing upon this line of study, Chen et al. (2021) utilised deep learning techniques to classify road networks across various global cities. By employing visual representations of street networks and deep convolutional neural networks, they categorised cities based on their morphological features. Importantly, this research established a link between these morphological categories and urban vitality, suggesting that cities with similar street network characteristics tend to exhibit comparable levels of urban activity. This relationship is critical for understanding how different urban forms impact city life, especially in the context of UK cities where variations in street patterns significantly influence urban dynamics.

Fleischmann et al. (2022) further expanded this field by proposing a numerical taxonomy of urban form, drawing inspiration from biological systematics. This taxonomy, which involves a detailed numerical characterisation of urban spaces including street networks, offers a systematic approach to classifying cities based on their spatial configurations. For the purposes of this study, such a taxonomy is particularly valuable as it allows for a rigorous comparison of UK cities based on their street network structures, independent of traditional metrics like size.

The dynamic nature of urban morphology is emphasised by Pili et al. (2017), who conducted a multi-criteria assessment of urban forms in the metropolitan region of Attica, Greece. This study categorised cities based on their morphological evolution, introducing a framework that accounts for changes in urban form over time. This dynamic perspective is essential for this research, suggesting that UK cities could be clustered not only by their current morphology but also by their developmental trajectories, offering a more comprehensive classification system.

Adding to this discourse, Webster (1995) introduced the concept of "urban morphological fingerprints," which describe the distinctive form of urban neighbourhoods through textural signatures of tessellated urban surfaces. This method enables the classification of cities based on the regularity and texture of their morphological patterns, offering a means to identify and categorise UK cities based on subtle, fine-grained differences in urban form.

Finally, Barthélemy (2017) provided new measures for analysing and categorising cities based on their street patterns. By examining the spatial distribution of betweenness centrality and the statistical properties of urban blocks, they classified cities into broad categories characterised by different abundances of block sizes and shapes. This method is particularly pertinent to the study of UK cities, where unique street network configurations can provide critical insights into how urban form influences city function and identity.

While significant advancements have been made to classify cities based on various morphological and street network characteristics, much of the existing research has been regional or has focused on specific cities, often integrating socio-economic factors. A comprehensive examination of UK cities, using street network topologies as the primary basis for classification, remains an area ripe for exploration. This research aims to fill this gap by offering a novel categorisation of UK cities based on their street networks, thereby contributing valuable insights to the fields of urban morphology and planning.

4.2 The Relationship Between Centrality and Retail Density

Understanding the relationship between centrality and retail density has been a central focus of urban studies, offering critical insights into how the structural characteristics of street networks influence commercial activity. This relationship was first brought to light in a foundational study by Porta et al. (2009), who analysed Bologna, Italy, using a Multiple Centrality Assessment (MCA) model. His findings revealed a significant positive correlation between global betweenness centrality and the concentration of retail activities. However, Bologna’s historical and compact street network, which likely enhances this correlation, raises questions about whether these findings can be applied to cities with different planning histories or urban forms. This indicates a gap in understanding how such relationships might vary in different urban contexts.

Building on these insights, Wang et al. (2014) extended the analysis to Changchun, China, further exploring the relationship between street centrality and retail location patterns. This study underscored the importance of context, revealing strong correlations for certain types of retail stores but also significant variability across different retail types. This variability highlights a critical gap in the literature: the relationship between centrality and retail density may not be universally strong across all urban environments, particularly in cities with distinct urban forms and socio-economic conditions, such as those in the UK. There is a clear need for research that explores this relationship in a wider range of cities with varying characteristics. In addition, Wang’s study uses a kernel density estimate (KDE) of centrality values rather than assessing individual street segments, which sacrifices the detailed insight that comes from analysing specific street-level interactions (Figure 5).

Liao et al. (2021) added another dimension by investigating the temporal dynamics of the relationship between centrality and retail density in Beijing. Introducing the concept of weighted centrality, which accounts for dynamic public transport flow networks, the study revealed how centrality’s influence on retail density shifts over time. However, Beijing’s dense public transport network and large-scale urban movement patterns suggest these findings may be highly context-dependent, limiting their applicability to other cities. This highlights the need for further research in different urban settings, particularly those with less integrated transportation systems or distinct developmental trajectories, where the relationship between centrality and retail density may differ significantly.

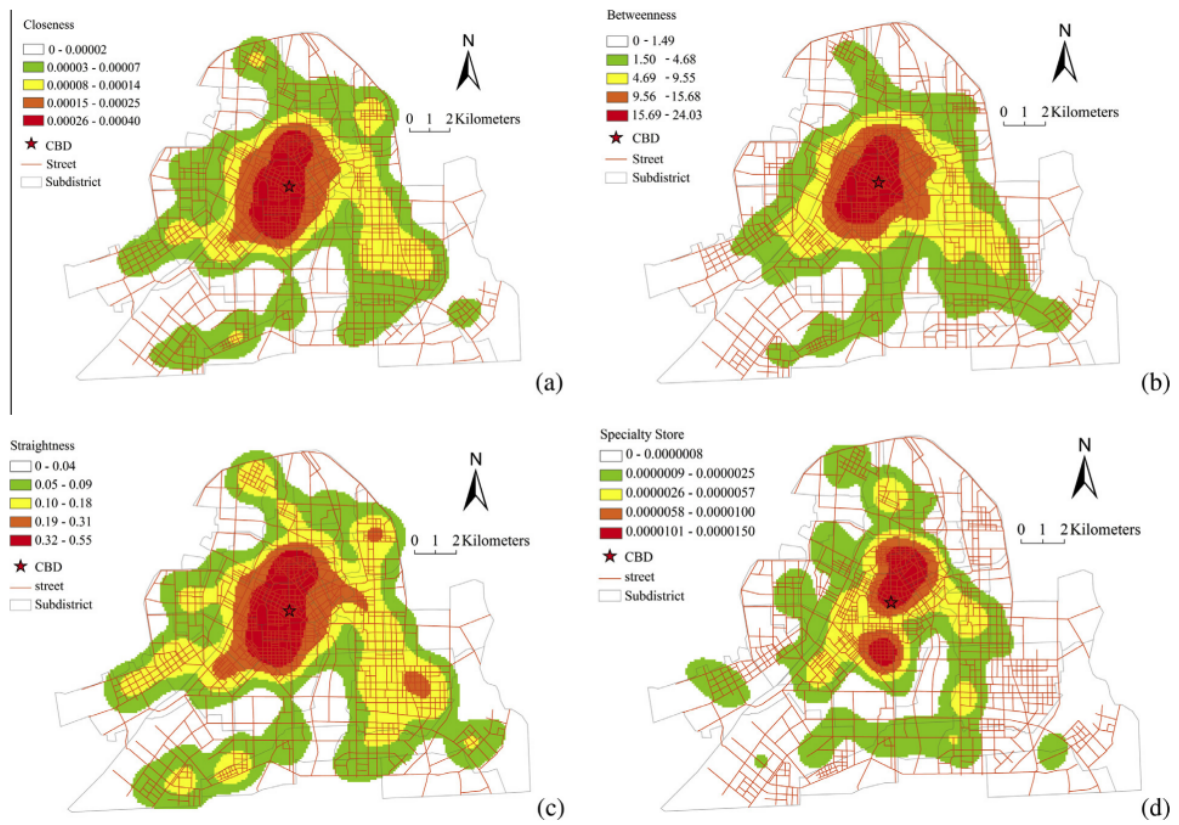


Figure 5: KDE values of (a) closeness, (b) betweenness (c) straightness, and (d) specialty stores (Wang et al. 2014)

Returning to the Bologna context, Strano et al. (2007) provided a more granular analysis by focusing on kernel density correlation (KDC) to examine the relationship between street centrality and commercial activities at the neighbourhood scale. While this approach offers detailed insights into how betweenness centrality influences retail density locally, the findings are again highly context-specific. Bologna’s well-preserved urban core may inherently favour a strong correlation between centrality and retail activity, but these results may not be generalisable to cities with different urban morphologies, such as those in the UK. This highlights a significant gap in the literature: the need for broader studies that examine how these relationships play out in cities with diverse urban forms.

Despite the valuable contributions of these studies, they collectively reveal a significant shortfall in the literature: the lack of comprehensive research that explores the relationship between betweenness centrality and retail density across a diverse range of urban contexts. The strong correlations observed in cities like Bologna and Changchun are deeply rooted in specific urban forms, dense street networks and unique retail landscapes. However, in cities with different characteristics, particularly those found in the UK, these relationships may not hold as strongly. This underscores the importance of conducting research that investigates this relationship across a broader spectrum of urban environments, including those with varying street network configurations and socioeconomic contexts.

Additionally, the methodologies employed in these studies, which often rely heavily on GIS-based analysis and kernel density estimation, introduce certain biases that can affect the outcomes. The choice of grid size, data resolution and the assumptions made during modelling can significantly impact the findings. Moreover, focusing exclusively on betweenness centrality may overlook other critical factors, such as economic policies, cultural preferences and the presence of anchor stores, all of which play essential roles in determining retail density. This methodological limitation points to the need for a more holistic approach in future research, one that integrates multiple factors to provide a more comprehensive understanding of retail location dynamics.

Finally, as highlighted by Liao et al. (2021), the relationship between centrality and retail density is not static but fluctuates over time, influenced by changes in transportation networks and urban development patterns. This temporal variability adds further complexity to the generalisation of findings across different contexts, especially in cities where retail activity is more dispersed or where transport networks do not align neatly with retail corridors. Consequently, while betweenness centrality can be a valuable tool in urban planning, there is a clear gap in the literature for studies that systematically explore its applicability across a wide variety of urban settings. This research addresses this gap by examining the relationship between street network topologies and retail density across an entire country—specifically, the UK—using a more nuanced and comprehensive approach that takes into account the diverse urban forms and contexts present within a single national framework.

4.3 Network Science, Space Syntax and Centrality

The application of network science concepts, particularly betweenness centrality, to urban systems has evolved significantly since the late 20th century, becoming a key tool in understanding the complexities of urban environments. The foundational work on network theory and centrality measures, including betweenness centrality, was pioneered by Linton C. Freeman in the late 1970s, most notably through his seminal paper, "A Set of Measures of Centrality Based on Betweenness" (Freeman 1978). This early work laid the groundwork for future studies, but it wasn't until the rise of GIS in the 1990s that the full potential of these concepts began to be realised. GIS enabled more sophisticated spatial analyses, facilitating the practical application of network theory to urban systems.

Space Syntax, developed by Hillier and colleagues in the 1970s, has been a foundational method for predicting and assessing retail density within urban environments. This approach provides a systematic framework for understanding how spatial configurations influence human behaviour and movement patterns, particularly in relation to accessibility and visibility within the urban grid (Hillier et al. 1976, Bafna 2003). By analysing the spatial morphology of cities, Space Syntax has been instrumental in identifying how street networks contribute to the concentration of retail activities and predicting areas of potential commercial success (Hillier 2012). While Space Syntax has offered valuable insights, its application has often been limited to specific case studies rather than comprehensive national assessments.

In contrast to these earlier methods, more recent studies have leveraged advanced network science concepts and GIS to enhance our understanding of urban environments. The integration of network theory, particularly betweenness centrality, into urban studies gained significant momentum in the 2000s. Michael Batty's influential work *Cities and Complexity* (Batty 2007, 2013) played a crucial role in bridging the gap between theoretical network science and practical urban analysis. Similarly, Carlo Ratti's application of Space Syntax principles through network analysis advanced our understanding of spatial configurations in cities, particularly in analysing pedestrian movement patterns. These contributions underscored the utility of network science in decoding the spatial dynamics of cities, although they also revealed a gap in the comprehensive application of these methods across diverse urban contexts.

Marc Barthélemy's work has furthered the application of network science in urban studies by introducing more complex mathematical models that explain the structural and topological properties of city networks. His research demonstrated how the physical layout of urban spaces influences functional dynamics, offering a deeper understanding of urban form and its relationship to city function. Building on these insights, my research extends Barthélemy's concepts to a national scale, applying them across the diverse urban contexts of UK cities. This approach not only tests the applicability of these models in a broader setting but also seeks to generalise the understanding of urban connectivity and its implications for planning and policy (Barthélemy 2011, Barthélemy 2017, 2018*a,b,c,d*).

By the 2010s, the use of network science in urban systems had become more widespread and sophisti-

cated, largely due to the availability of large datasets and advanced computational methods. Key studies from this period, such as those by Porta, Crucitti and Latora (Porta et al. 2009), analysed the urban street networks of different cities to understand their structural properties, further cementing the role of network science in urban studies. Concurrently, the work of Barabási and colleagues on "scale-free networks" introduced new dimensions to the understanding of urban systems, highlighting the non-uniformity and complexity of city structures.

Despite these advancements, the application of network science, particularly betweenness centrality, has predominantly been examined in a limited range of urban settings. While these studies have provided valuable insights into the connectivity, accessibility and spatial dynamics of urban systems, there remains a significant gap in the literature: the need for a more comprehensive application of network science across a broader spectrum of cities, particularly in understanding how these concepts can be applied at a national scale. Most existing research has focused on individual cities or specific regions, often neglecting the diversity and complexity found within an entire country.

My research seeks to address this gap by applying network science and betweenness centrality to the categorisation of UK cities based on their street network topologies. This approach not only provides a more nuanced understanding of how these cities are connected and structured but also offers insights into how network science can be used to analyse urban systems at a national level. By extending the application of network science beyond individual case studies to a comprehensive analysis of an entire country, this study aims to contribute to a more robust and generalisable understanding of urban connectivity and its implications for urban planning and policy.

5 Methodology

5.1 Ethical Considerations

The level of ethical risk in this investigation is considered minimal. This is because all data used in this study, aside from Ordnance Survey (OS) *Points of Interest*, is freely available to the public. OS *Points of Interest*, which is available for purchase, has been accessed through an educational license via the EDINA DigiMap.

This study ensures that none of the licensed data will be published or shared in a manner that could lead to unauthorised use or risk of misuse. All analysis is conducted within the bounds of the educational license agreement and results are presented in an aggregated form to prevent the disclosure of sensitive or proprietary information.

5.2 Data Collection and Wrangling

The datasets used in this study were obtained from the OS, the national mapping agency for Great Britain. OS carries out the official surveying of GB, providing accurate geographic data for public use (DSIT 2017). These datasets and the preprocessing required are outlined below. Data wrangling and analysis were conducted using Python, C++ and QGIS.

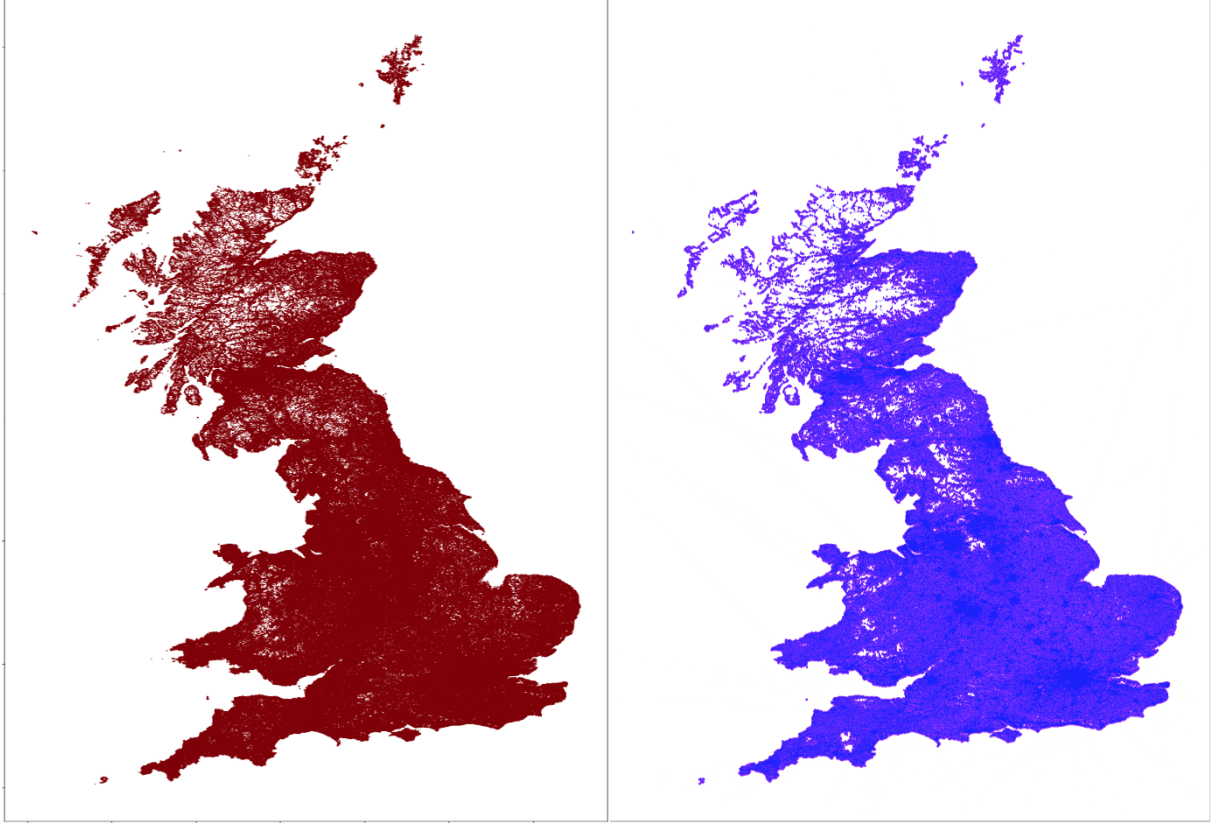


Figure 6: Visualisation of Raw Ordnance Survey Points of Interest (Left) and OpenRoads (Right) Datasets

5.2.1 OS OpenRoads Data

The OS OpenRoads dataset is a comprehensive representation of Great Britain’s motorways, A-roads, B-roads, road classifications, road names, primary route information and motorway junction information. Updated twice a year, this dataset is available as vector points and lines. It includes approximately three million intersections and four million road segment in its raw form. To ensure the information used in this study is current we download the latest version released in 2024 as a GeoPackage. Figure 6 illustrates the nodes in purple and roads in blue. This data was processed in 2 ways. For computational purposes the size of the graph is reduced by removing the attributes of intersections and streets ”...unnecessary to our analysis and points that do not convey any morphological information, such as nodes of degree two, which for example correspond to streets changing name; and by representing roundabouts by a single intersection point” (Arcaute et al. 2016, Simons 2023). Secondly, this study requires the calculation of

complex network metrics including degree, closeness and betweenness centrality. These metrics require data to be projected as a graph. Due to the speed of the Python library *graph_tool*, OpenRoads was transformed into a *gt graph* (Appendix B).

5.2.2 OS Points of Interest Data

The second dataset is OS *Points of Interest* (POI), which contains over four million records classified in a three-tier hierarchy comprising nine groups, 52 categories and 616 classes (Survey 2024b). Each record includes a unique reference number, feature name, classification code, geo-code, positional accuracy code, topographic reference, distance to the nearest road and provenance reference (Hewitt 2024, Survey 2024a). Updated quarterly, the POI data are separated into urban and non-urban settings to account for differences in behavioural patterns. For analysis, the POI data is examined at the 52-category level, providing a balance between the over-generalised nine themes level and the overly detailed 616 classes level (Hewitt 2024, Survey 2024a). Figure 6 we see each observation in the complete dataset represented by red point data. Before conducting analysis, OS POI was filtered on exclusively retail locations, this reduced the observation count from approximately 4 000 000 to around 326 000.

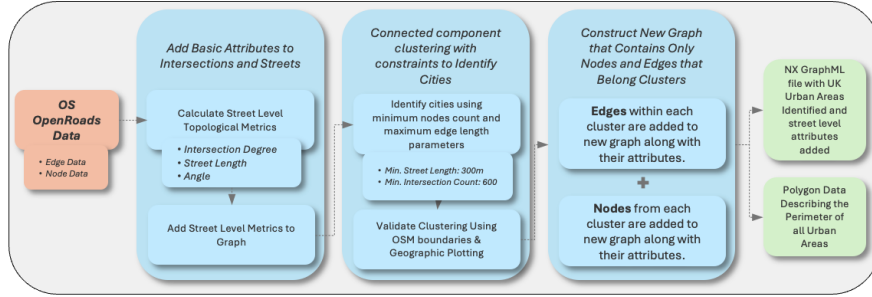
5.2.3 OSM Geographical City Perimeters

To verify the clustering of street networks into cities, the Python library OSMnx was used to obtain the official boundaries of UK cities from OSM. These boundaries were then cross-referenced with the urban areas identified through clustering. This validation step ensured that the clustered cities matched legitimate city boundaries in the UK. This step is visualised below in Figure 11.

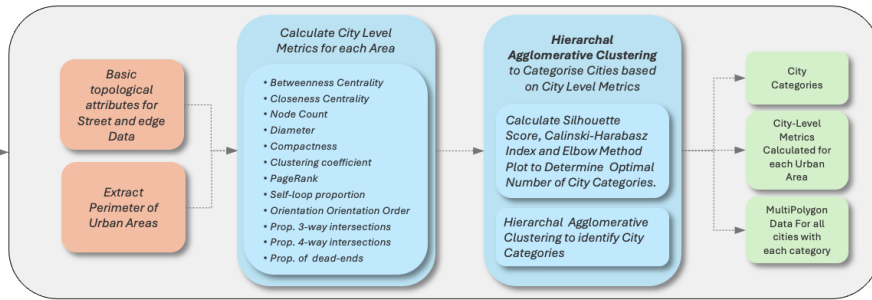
5.3 Workflow and Methods

An array of spatial, statistical and machine learning methods were combined into a seamless workflow to conduct the analysis. Figure 7 illustrates the workflow diagrammatically and explanations of individual stages are detailed in the following paragraphs.

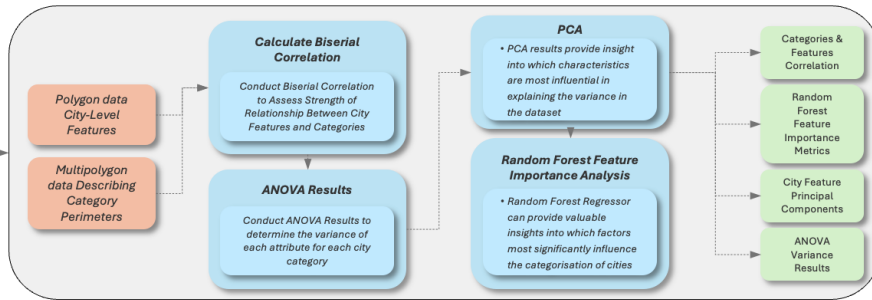
Stage 1: Conducting Unsupervised Clustering to Identify UK Cities



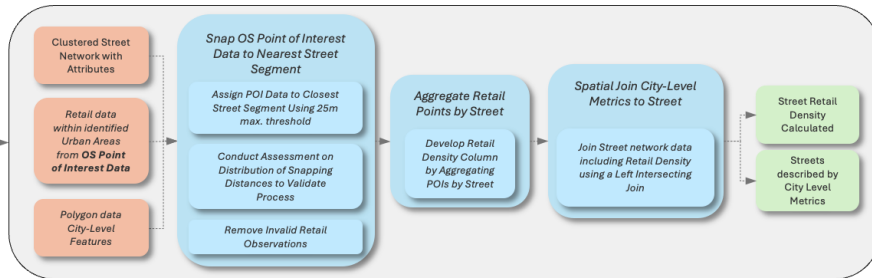
Stage 2: Categorising UK Cities Using City Level Metrics



Stage 3: Investigate Clusters, Principal Components and Feature Importance



Stage 4: Conduct Spatial Join Between Retail Density Data & Street Network



Stage 5: Assess Relationship Between Retail Density & Street Attributes at Different Scales

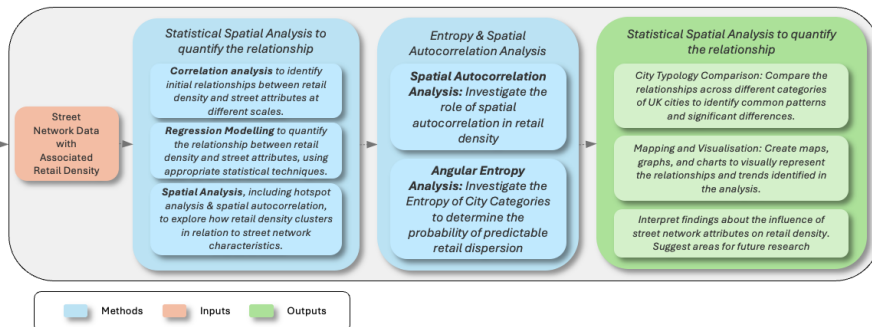


Figure 7: Overview of Entire Workflow

5.3.1 Stage 1: Conducting Unsupervised Clustering to Identify UK Cities

To topologically define UK cities, this study builds on the research of Arcaute et al. 2016. Their hierarchical percolation study demonstrates that UK cities exhibit a maximum fractal dimension when the maximum road segment length is set to 300m, effectively capturing the complexity and intricacies of urban spatial structures (2016). Additionally, their study establishes that meaningful urban clusters are best defined using a minimum node count of 600, providing a clear threshold for capturing the essential connectivity and structure of urban networks (2016). For efficiency, this study employs **connected component clustering** using these parameters, a method that identifies distinct subgraphs within a larger graph, where each subgraph (or cluster) consists of interconnected nodes that have no connections to nodes in other subgraphs (Newman 2010).

As street length is not an attribute included in *OS OpenRoads* we calculate street length using the geometry column. In addition we add several other street level metrics which can be calculated without contextualising it by the city it is in, these include node degree, the number of edges (streets) connected to each node (intersection) and node angle, which is calculated by determining the average angle between all pairs of neighboring nodes, using an arctangent function to compute the angle between vectors originating from the node to each of its neighbors. These equations are outlined below (Boeing 2022).

Degree Centrality Degree Centrality measures the number of direct connections a node has, highlighting stations that serve as major hubs for passenger transfers (Stamos 2023, Napitupulu et al. 2021). The non-normalised degree centrality of a node v in a network is given by the sum of its adjacency matrix entries, where $a_{v,i}$ is 1 if there is an edge between node v and node i and 0 otherwise.

$$deg(v) = \sum_{i=1}^N a_{v,i} \quad (1)$$

N represents the total number of intersections or nodes in the network. v represents an intersection, with $deg(v)$ indicating the number of street segments connected to that intersection, normalised by $n - 1$ (Stamos 2023). The normalised degree is thus:

$$C_D(v) = deg(v) \quad (2)$$

Node Angle Node angle is calculated by determining the average angle between all pairs of neighbouring nodes. This is done using an arctangent function to compute the angle between vectors originating from the node to each of its neighbours (Mortensen 2020). Let $N(v)$ represent the set of neighbours of node v . The angle θ_{ij} between vectors from v to its neighbours i and j is calculated using the arctangent function, where x_i, y_i and x_j, y_j are the coordinates of neighbours i and j respectively (Castro 2013, Tandeitnik 2023).

$$\theta_{ij} = \arctan \left(\frac{(y_j - y_v)(x_i - x_v) - (y_i - y_v)(x_j - x_v)}{(x_i - x_v)(x_j - x_v) + (y_i - y_v)(y_j - y_v)} \right) \quad (3)$$

The average node angle $\bar{\theta}(v)$ for node v is given by the mean of all such angles between pairs of neighbours:

$$\bar{\theta}(v) = \frac{2}{|N(v)|(|N(v)| - 1)} \sum_{i,j \in N(v), i < j} \theta_{ij} \quad (4)$$

where $|N(v)|$ is the number of neighbours of node v .

5.3.2 Stage 2: Calculating City Level Metrics and Categorising UK Cities

Geoff Boeing’s street network models and indicators offer an excellent framework for analysing urban areas due to their methodological rigor, global applicability and reproducibility. His metrics, derived from OSM data using the OSMnx toolkit, allow for consistent and scalable analysis across urban contexts. These metrics not only address critical challenges related to data accessibility and consistency but also provide robust insights into the topological characteristics of street networks. Consequently, applying Boeing’s *topological* metrics in this study is an excellent approach to effectively categorise and compare UK cities based on their spatial organisation and connectivity (Boeing 2019, 2022). These metrics are defined, below, in Table 1.

Term	Definition
Closeness Centrality	Measures how easily accessible a particular intersection is from all other intersections within a defined radius.
Clustering Coefficient	Indicates the degree to which nodes in a graph tend to cluster together, reflecting the density of triangular connections in the network.
Betweenness Centrality	Quantifies the importance of intersections in facilitating movement through the street network by measuring the number of shortest paths that pass through each intersection.
Average Node Degree	Calculates the average connectivity of intersections, providing an overall sense of network complexity.
Proportion of 4-way Intersections	The ratio of intersections with four connecting streets, which is indicative of a grid-like urban structure.
Proportion of Dead-ends	The ratio of dead-end streets, which impacts the navigability and connectivity of the network.
Orientation Entropy	Measures the disorder or randomness in the orientation of street segments, indicating the complexity of the street layout.
Orientation Order	Quantifies the alignment and uniformity in the orientation of street segments, reflecting the grid-like structure of the street network.
Self-loop Proportion	The ratio of streets that loop back to the same intersection, affecting the overall connectivity and complexity of the network.
PageRank	Assesses the importance of intersections based on the connectivity and the number of connections each intersection has, similar to Google’s PageRank algorithm.

Table 1: Geoff Boeing’s City-Level Topological Metrics and Their Definitions (Boeing 2019, 2022)

In the following section, Boeing’s metrics and their calculations are explained in more detail.

Betweenness Centrality Betweenness measures the extent to which a node lies on the shortest path between other nodes in the network. In the context of the London street network, it identifies intersections through which the highest number of shortest paths pass through, indicating intersections that are critical for maintaining connectivity across the network (Derrible 2012, Stamos 2023).

$$C_B(v) = \sum_{s,t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (5)$$

$\sigma_{st}(v)$ represents of the number of shortest paths from node s to node t that traverse node v . σ_{st} signifies the number of shortest paths between s and t . In the context of the London street network, v denotes an intersection and $\sigma_{st}(v)$ indicates the number of routes between other intersections s and t that include intersection v (Stamos 2023).

This equation can be normalised using N to represent the total number of nodes in the network (Stamos 2023, Ye et al. 2020):

$$C_B(v) = \frac{1}{N^2} \sum_{s,t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (6)$$

Closeness Centrality Closeness is a score that indicates how physically close a node is to all other nodes in the network based on edge length. It identifies intersections that offer the shortest (distance) average journey time to all other intersections (Stamos 2023).

$$C_i = \frac{1}{l_i} = \frac{n}{\sum_j d_{ij}} \quad (7)$$

l_i represents the average shortest path length from node i to all other nodes and d_{ij} signifies the shortest path length between nodes i and j and n again is the total number of intersections. In the context of the the London street network, i represents an intersections and l_i indicating the average distance from intersection i to all other intersections calculated as the reciprocal of the sum of shortest path lengths (Stamos 2023).

Orientation Entropy Orientation Entropy measures the disorder or randomness in the orientation of street segments within a network, reflecting the complexity and variation in street layout. It is calculated using Shannon’s entropy formula, where higher values indicate a more disordered or irregular street network (Boeing 2019).

$$H = - \sum_{i=1}^k p_i \log(p_i) \quad (8)$$

Here, p_i represents the proportion of street segments oriented in direction i and k is the total number of unique orientations present in the network (Boeing 2019).

Orientation Order Orientation Order quantifies the alignment and uniformity in the orientation of street segments, indicating the degree to which a street network follows a grid-like structure. It is typically calculated as the inverse of orientation entropy (Boeing 2019) .

$$O = \frac{1}{H} \quad (9)$$

In this equation, O is the orientation order and H is the orientation entropy. Lower values of H (higher alignment) result in higher values of O , reflecting a more ordered and grid-like network (Boeing 2019).

Self-loop Proportion Self-loop Proportion is the ratio of streets that loop back to the same intersection in the network, influencing the overall connectivity and complexity. It is calculated as follows:

$$S = \frac{\text{Number of Self-loops}}{\text{Total Number of Streets}} \quad (10)$$

In this equation, S denotes the self-loop proportion. A higher value indicates a greater proportion of streets that form loops, affecting navigability and network redundancy (Boeing 2019).

PageRank PageRank assesses the importance of intersections within a street network based on their connectivity and the number of connections each intersection has. It is derived from Google’s PageRank algorithm and can be expressed as (Boeing 2019):

$$PR(v) = \frac{1-d}{N} + d \sum_{u \in M(v)} \frac{PR(u)}{L(u)} \quad (11)$$

Here, $PR(v)$ is the PageRank of intersection v , d is the damping factor (typically set to 0.85), N is the total number of intersections, $M(v)$ represents the set of intersections that link to v and $L(u)$ is the number of links from intersection u . This metric identifies key intersections that significantly influence the network’s connectivity (Boeing 2019).

To categorise the cities, we employ hierarchical agglomerative clustering, a method that builds nested clusters by progressively merging or splitting them based on a measure of similarity. This approach is advantageous over other forms of clustering, such as k-means, because it does not require specifying the number of clusters in advance and can reveal the hierarchical structure of the data.

Although hierarchical clustering does not necessitate predefining the number of clusters, we add certainty to our clustering by determining the optimal number of clusters using the *Silhouette Score*,

Calinski-Harabasz index and an *Elbow Method* plot. These metrics assess the quality and coherence of the clusters. This clustering technique groups cities with similar topological characteristics, enabling us to identify distinct urban typologies and understand their spatial organisation.

5.3.3 Stage 3: Investigate Clusters and Feature Importance

An examination of our city categories will identify underlying patterns and relationships within urban street networks, which is essential to understand the varying scales and structural features inherent in each category and articulates its specific topological characteristics.

To explore this relationship between topological metrics and the identified clusters, we calculated biserial correlation coefficients—statistical measures that assess the relationship between a continuous variable and a binary variable with underlying continuity—for each metric across the clusters. This allowed us to quantify the strength and direction of associations between the metrics and clusters, providing insights into which topological features were most strongly associated with each cluster (Kornbrot 2014).

We also employed a Random Forest model to determine the importance of each topological feature in distinguishing between the clusters. Feature importance in a Random Forest model quantifies the impact of each feature on the model's predictive accuracy, assessing its contribution across all decision trees. This analysis helps to understand the critical role of various topological metrics in distinguishing city categories (Menze et al. 2009).

To identify the most important features that define the categories of cities in our study, we performed Principal Component Analysis (PCA). PCA was used not just for dimensionality reduction but also to determine which topological metrics are most influential in distinguishing different categories of UK cities. By examining the loadings of each topological metric on the principal components, we gained insights into the primary dimensions that are key to differentiating these urban areas.

5.3.4 Stage 4: Snap Retail Density Data to Street Network

In this step, the objective was to conduct a spatial join of the retail POIs data with the street network data in order to analyse the relationship between retail density and street network attributes. The process began by reading and preparing the retail data from a GeoPackage file. The data was re-projected to the British National Grid (EPSG:27700) to ensure consistency with the street network data. The geometry of the retail points was defined using their easting and northing coordinates and unnecessary columns were removed to retain only relevant attributes such as 'groupname', 'categoryname', 'pointx_class' and 'classname'.

The retail data was then filtered to include only relevant categories *Eating and Drinking* and *Retail Services*. This filtered data was visualised to ensure its accuracy and relevance to the study. Additionally, the entire dataset was filtered to retain only points within the identified urban areas, ensuring that only

relevant POIs were included in the subsequent analyses. This step involved checking each point to see if it fell within any city polygon and keeping only those that did. The filtered retail dataset is visualised in Figure 8

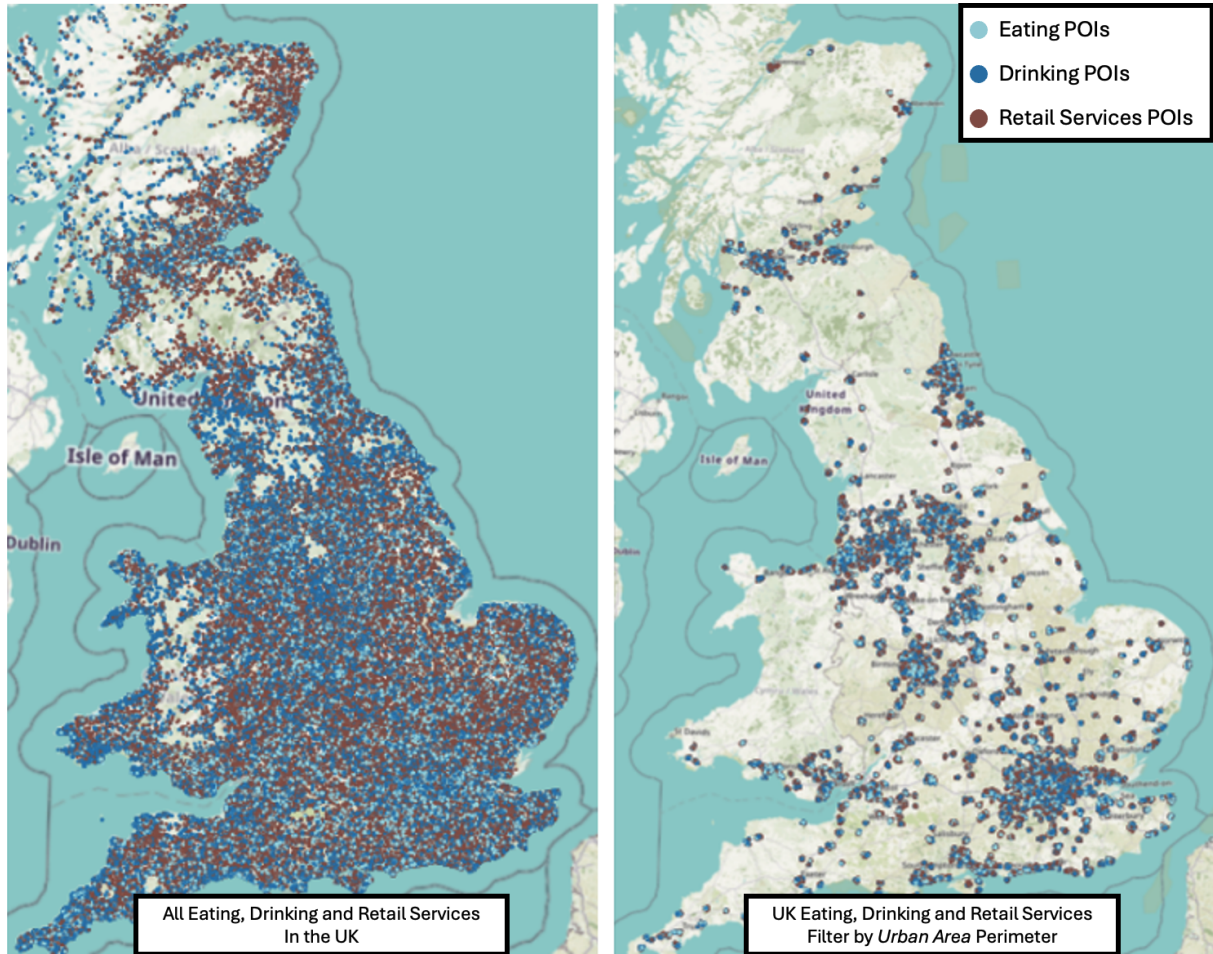


Figure 8: The left map shows all eating, drinking, and retail service POIs across the UK, while the right map filters these POIs to urban areas only. Eating POIs are in teal, drinking POIs in blue, and retail services POIs in red, illustrating their distribution and concentration in both national and urban contexts.

POIs were snapped to the nearest road segment, this, in essence associated each POI with a street edge. A maximum snapping distance of 25 meters was set to maintain meaningful associations between POIs and street segments. In Figure 9 we see a Kernel Density Estimate (KDE) plot and a histogram of snapping distances. Both plots support a 25-meter maximum snapping distance by showing that the majority of snapping distances fall between 10 to 20 meters, with a sharp decline beyond 20 meters. This indicates that a 25-meter threshold effectively captures most relevant points without introducing excessive noise, ensuring the precision and accuracy of the spatial relationships in the analysis (ESRI 2024) .

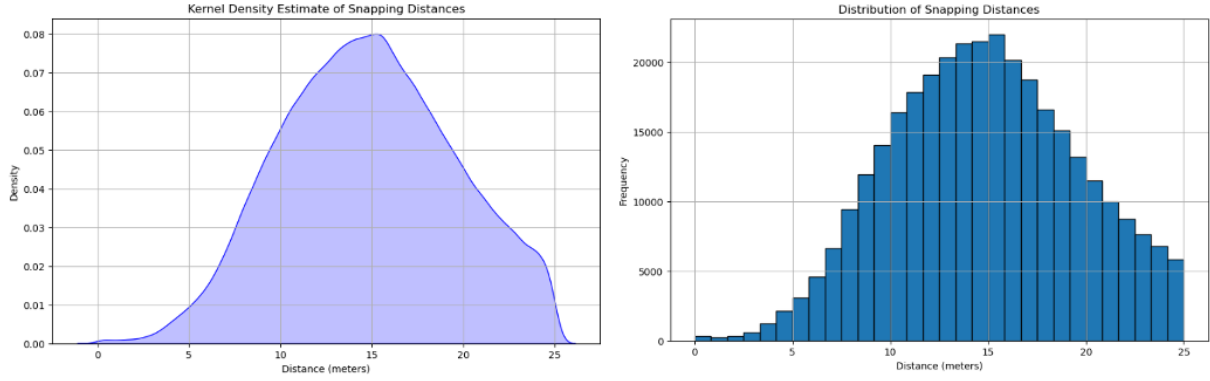


Figure 9: Kernel Density Estimate and Raw Distribution of Snapping Distances

Finally, a spatial join was performed to associate each POI with its respective nearest road segment, following the process outlined above. This allowed for the aggregation of retail density metrics by street segment, facilitating a detailed analysis of the relationship between retail density and various topological metrics of the street network. The spatial join process involved merging the attributes of POIs with the corresponding road segments based on their spatial proximity and verifying the integrity of the joined dataset. This comprehensive approach ensured the robustness of the spatial join and the reliability of the resulting data for analysis.

5.3.5 Stage 5: Assess Relationship Between Retail Density and Street Attributes at Different Scales

In this step, we aimed to explore how different categories of UK cities, defined by their street network topologies, exhibit distinct patterns in retail density and distribution and to identify which topological metrics are most strongly correlated with retail density across various categories of UK cities. The following techniques were employed to assess these relationships: Random Forest Regressor, Decision Tree, Linear Regression and conventional correlation analyses.

Firstly, the street network data, including the retail density metrics previously joined, was segmented into various categories based on hierarchical clustering results from earlier stages. The observations were grouped into distinct city categories to facilitate a detailed comparison of retail density patterns. This segmentation allowed for the identification of variations in retail density distribution within each city category.

To quantify the relationship between retail density and street network attributes, statistical spatial analysis was conducted. Pearson correlation analysis was performed to evaluate the strength and direction of associations between retail density and various topological metrics, including node count, clustering coefficient and betweenness centrality. This provided quantitative insights into which topological metrics were most strongly associated with retail density. The findings showed weak or no significant correlations across the different categories, suggesting that these metrics alone do not robustly predict retail density.

To better understand the weak relationship between retail density and centrality, it is crucial to account for the spatial patterns and variability within the urban street network. Spatial autocorrelation and entropy are employed as they provide insights into the extent of spatial clustering and the complexity of street orientations, both of which can significantly influence the distribution of retail locations (Lee 2017, Marin et al. 2022).

To assess spatial autocorrelation, we calculated Moran’s I for retail density using the weights matrix derived from the street network data. Moran’s I measures the degree to which similar values of retail density are clustered together in space, indicating whether the retail density distribution is random or exhibits spatial clustering. If the results do show significant spatial autocorrelation, this would suggest that retail locations are not randomly distributed but tend to cluster in specific areas. This clustering can obscure the influence of topological metrics on retail density, as nearby locations may have similar retail densities due to external factors such as socioeconomic conditions or planning regulations (Lee 2017).

Furthermore, we will calculate Shannon’s entropy for each city category to measure the randomness and disorder of street orientations. Higher entropy values indicate a more complex and less predictable street network, which could influence the distribution of retail locations. High Shannon’s entropy levels, reflect the complexity and variability in their street networks. This complexity may contribute to the weak correlations observed between topological metrics and retail density, as the street network’s structural randomness can dilute the predictive power of these metrics (Marin et al. 2022).

These analyses underscore that while topological metrics provide valuable insights into the structural characteristics of urban areas, other factors such as spatial clustering, population density and socioeconomic variables also play significant roles in shaping retail density patterns. Integrating these factors into future models could enhance the predictive power and provide a more comprehensive understanding of the determinants of retail density in urban areas.

6 Analysis and Results

6.1 Analysis

We defined our system of cities by conducting unsupervised clustering on OS OpenRoads Data, following the parameters outlined in Arcaute et al. 2016. This approach allows us to move beyond traditional administrative boundaries and define urban areas based on the intrinsic connectivity of the street network. The result is the identification of 534 urban areas across the UK, as depicted in Figure 10. Notably, Manchester and Liverpool have been clustered together, likely due to significant expansions in the street network since Arcaute et al.’s 2016 study.

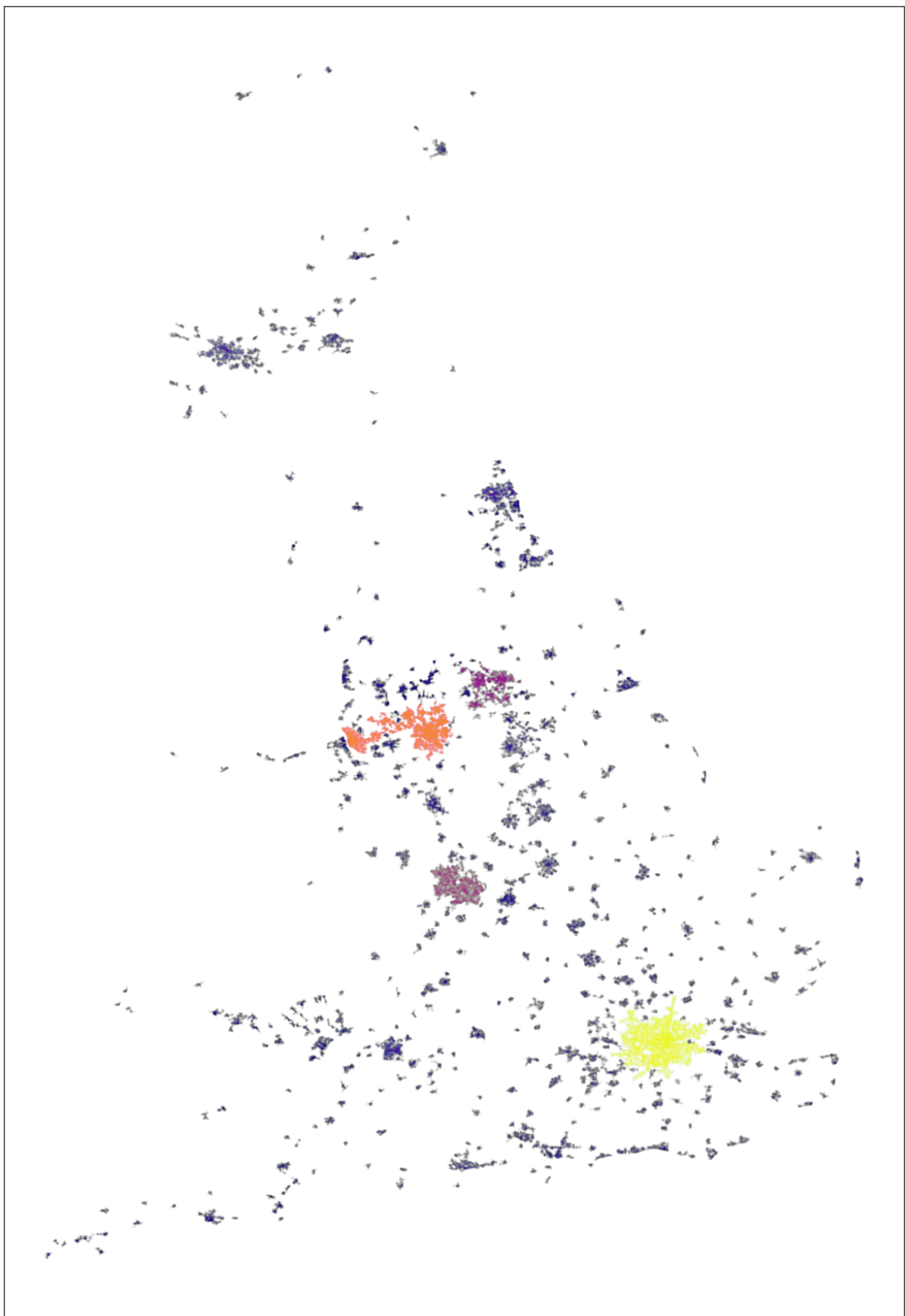


Figure 10: **Visualisation of 534 Urban Areas Identified by Clustering.** Each urban area is color-coded by size, with lighter colors representing larger clusters, highlighting the distribution and relative scale of urban clusters across the UK. Each point represents a street segment.

After applying the connected component clustering to the street network graph, the resulting clusters were evaluated for geographic and administrative validity. Each identified cluster corresponded closely to known city boundaries, as verified by overlaying the clusters on a detailed map of the region obtained from OSM (2024). Figure 11 illustrates the perimeters of the UK's 500 largest urban centres alongside the clustered street segments defined in this study, demonstrating the effectiveness of our approach with a 96.31% overlap ¹.

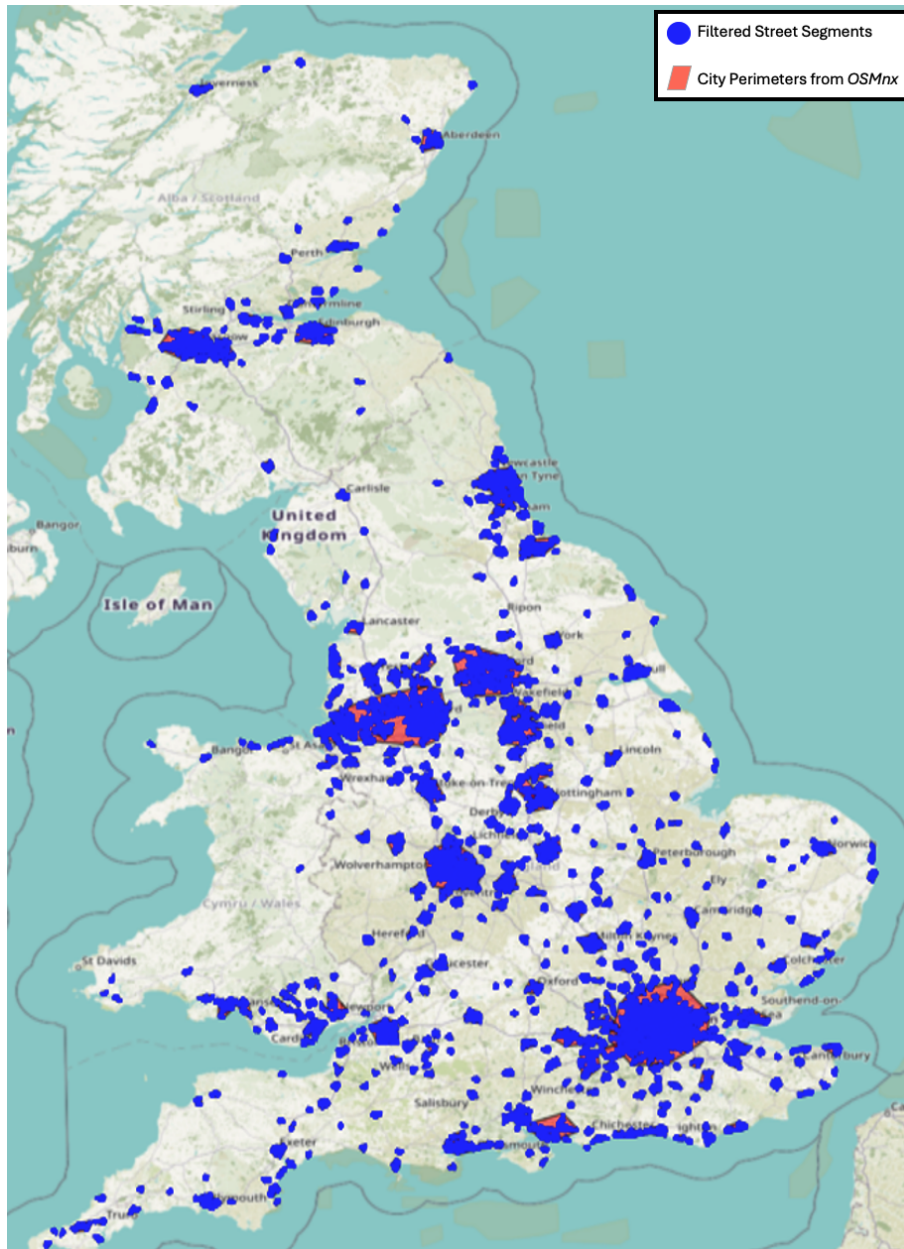


Figure 11: Distribution of Street Segments Within the 534 Largest UK Cities. Blue represents all street segments, with 96.31% lying within the city perimeters (orange) as defined by OSM, highlighting the concentration of urban infrastructure.

At this point in our analysis, we have successfully defined and validated the Urban Areas within the

¹The Perimeter of Manchester and Liverpool has been combined to ensure alignment with identified clusters

UK. In order to identify distinct patterns in retail density and distribution across different categories of UK cities, it is necessary to develop a taxonomy of UK cities based on their topological network. We leverage Silhouette Scores, the Elbow Method and the Calinski-Harabasz Index to find the optimal number of clusters is 11. These results are visualised in Figure 12.

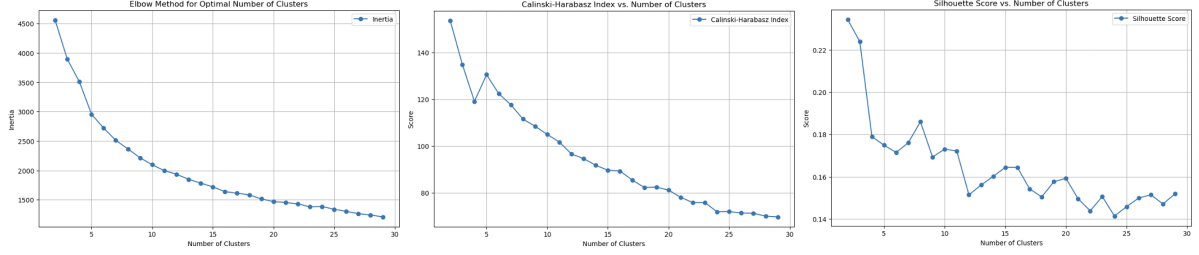


Figure 12: Silhouette Score, Calinski-Harabasz Index and Elbow Method Scores Plotted

Clustering reveals the 11 following city categories. Figure 13 visualises the geographical distribution of road edges by city category. Initial inspection reveals that hierarchical clustering effectively highlights various urban structures and boundaries that align with complex geographical, political and historical processes in the UK (Arcaute et al. 2016).

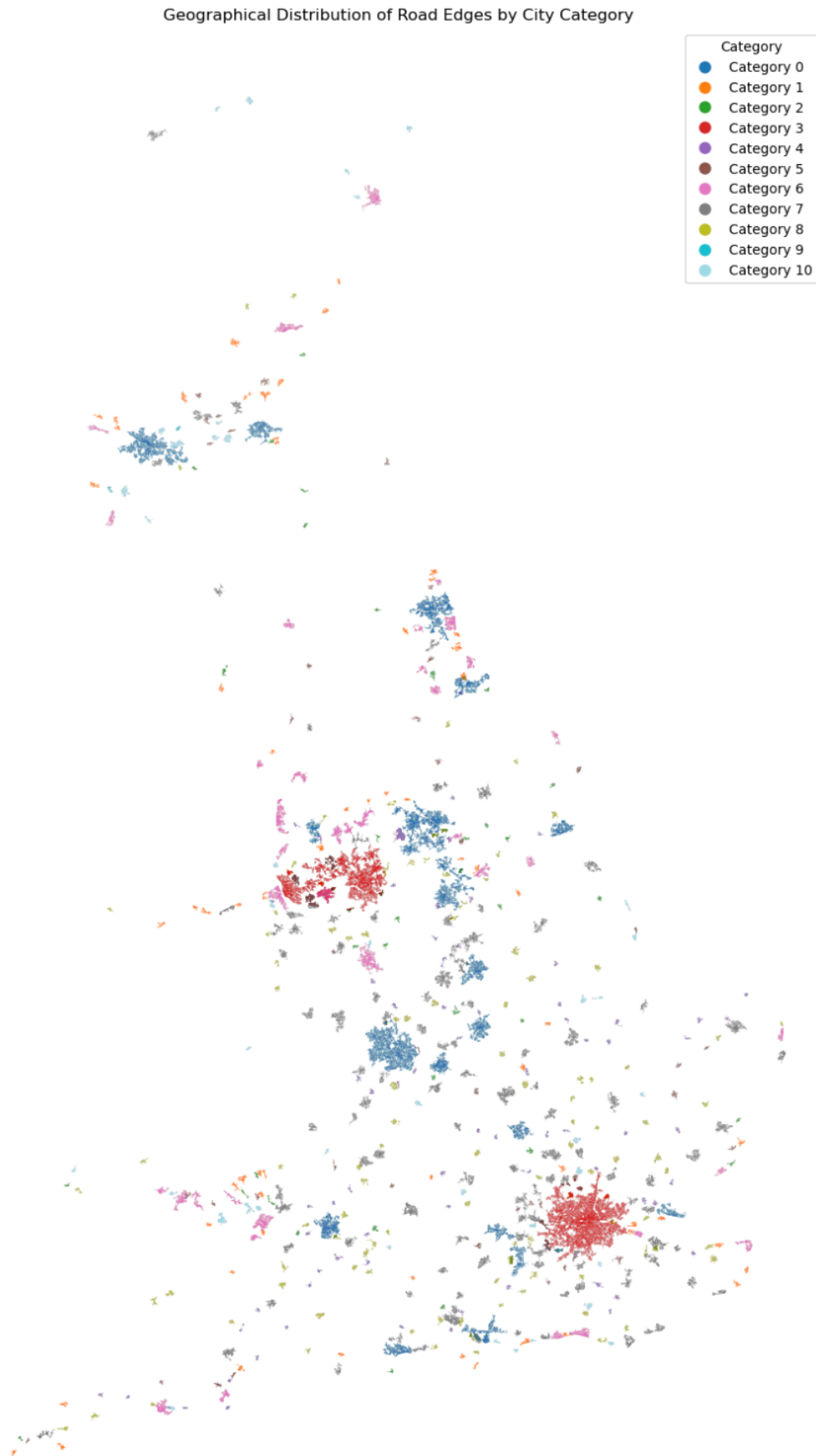


Figure 13: UK cities categorised by topological metrics, with each colour representing a distinct category.

At this point, we have successfully categorised UK cities based on their topological metrics. We now investigate the topological nature of these categories to better understand their hierarchical structure, connectivity and functional roles.

Analysis of Variance (ANOVA) Results An ANOVA test is a statistical method used to determine if there are significant differences between the means of three or more groups, making it particularly useful for comparing multiple categories within a dataset to understand if they differ in a meaningful way. The ANOVA results reveal significant differences in the topological street network characteristics across UK city categories. Metrics such as diameter, clustering coefficients and intersection proportions (3-way, 4-way and dead-ends) exhibit substantial variability, as evidenced by high F-statistics and extremely low p-values. For instance, the diameter metric shows a significant difference with an F-statistic of 257.70 (p-value = 7.64×10^{-195}). Similarly, *Maximum PageRank* and orientation metrics also display significant differences, underscoring the diversity in urban forms and their implications for city planning and commercial activities. These results highlight the importance of considering these structural variations in urban analysis and strategy development.

Metric	F-statistic	p-value
Diameter	257.70	7.64e-195
CC Avg Dir Value	90.52	2.69e-107
CC Avg Undir Value	90.52	2.69e-107
Proportion 3-way	72.82	2.02e-92
Proportion 4-way	56.32	1.48e-76
Proportion Dead-end	91.87	2.35e-108
PageRank Max	139.05	2.79e-140
Self-loop Proportion	45.70	4.64e-65
Orientation Entropy	61.13	2.08e-81
Orientation Order	226.22	1.01e-182

Table 2: ANOVA Results for UK City Categories Based on Street Network Topology

Biserial Correlation Analysis Building upon the biserial correlation analysis, the results presented in Figure 14 further illustrate the intricate relationships between topological metrics and the urban clusters identified in the study. For example, Cluster 10 shows strong positive correlations with orientation order and PageRank Max, indicating that cities within this cluster exhibit a high degree of street alignment and centrality, characteristics often associated with systematically planned urban environments. Similarly, Cluster 2, with notable positive correlations with orientation order and clustering coefficient, reflects a high regularity and connectivity in street layouts, typical of highly organised urban settings.

On the other hand, clusters like 4 and 6, which display significant positive correlations with PageRank Max and proportion of 4-way intersections, suggest the presence of structured, grid-like patterns and centrality, often found in well-planned urban grids. Meanwhile, clusters such as 8 and 3, which have weaker correlations across most metrics, indicate a more varied and less predictable urban form, likely

representing a mix of organic and planned growth.

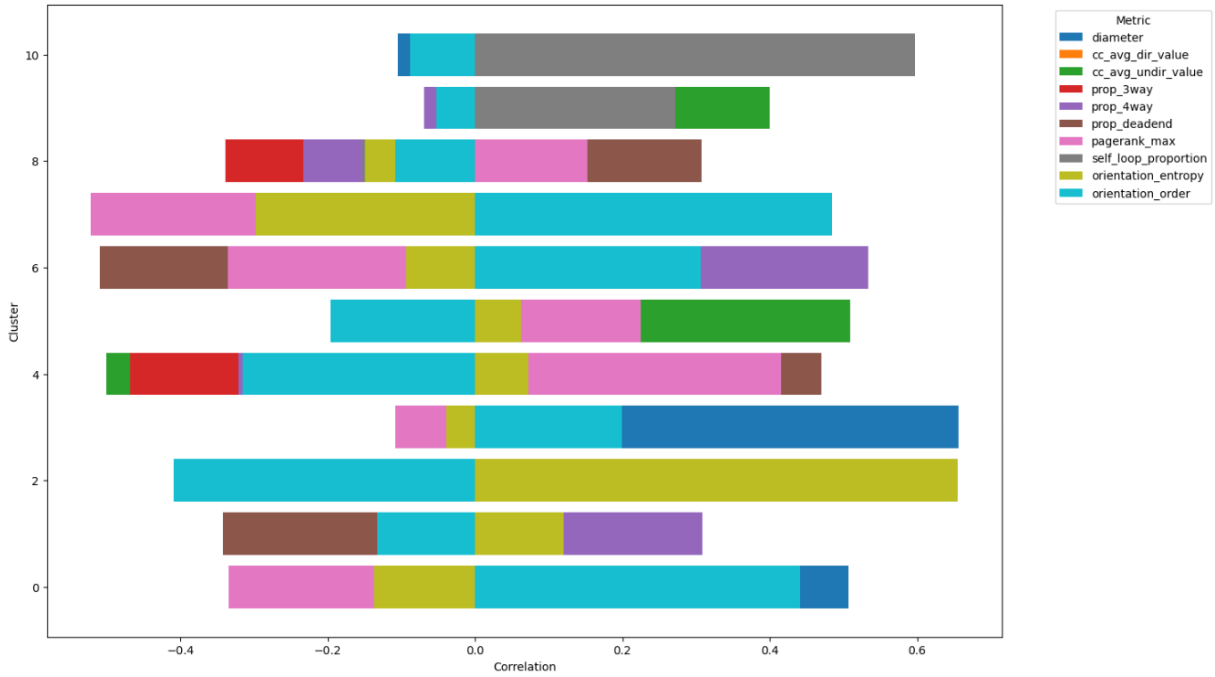


Figure 14: Biserial Correlation Between Topological Metrics and Clusters

6.1.1 Cluster 3 (Red): Largest Cities - London, Manchester and Liverpool

Cluster 3, represented in red, includes the largest cities in the UK, such as London, Manchester, and Liverpool. These cities are highly correlated with diameter and orientation order, indicating that their extensive and well-structured street networks support significant urban sprawl and organised layouts. The high correlation with diameter underscores the vastness of these cities, while the correlation with orientation order reflects the systematic planning that contributes to their prominence as central hubs in the UK. These cities play a critical role in the national street network, facilitating movement and connectivity across large urban areas.

6.1.2 Cluster 0 (Blue): Second Largest Cities - Birmingham, Edinburgh, Glasgow, Nottingham, Leeds, Bristol

Cluster 0, depicted in blue, includes major regional hubs such as Birmingham, Edinburgh, Glasgow, Nottingham, Leeds, and Bristol. These cities exhibit strong correlations with orientation order, orientation entropy, and PageRank Max, highlighting their complex and well-organised street networks. The correlation with orientation order and entropy suggests a high degree of alignment and regularity in their street layouts, while the PageRank Max correlation indicates their central role in regional connectivity. Although these cities are significant, their networks are slightly less expansive compared to Cluster 3, reflecting their status as key regional centres rather than national hubs.

6.1.3 Cluster 6 (Pink): Third Largest Cities - Blackpool, Cardiff, Worthing, Plymouth, Torquay

Cluster 6, represented in pink, includes third-tier cities such as Blackpool, Cardiff, Worthing, Plymouth, and Torquay. This cluster shows significant positive correlations with orientation order, orientation entropy, PageRank Max, and the proportion of dead ends. The strong correlation with orientation order and entropy indicates a systematic yet varied street layout, while the positive correlation with the proportion of dead ends suggests a mix of planned and organically grown areas. These cities serve as important regional centres, with street networks that are organised yet less complex than those of larger urban areas.

6.1.4 Cluster 7 (Grey): Fourth Largest Cities - Oxford, Eastbourne, York, Grimsby, Lincoln

Cluster 7, shown in grey, includes cities like Oxford, Eastbourne, York, Grimsby, and Lincoln. These cities are characterised by moderate correlations with orientation order, orientation entropy, and PageRank Max, indicating a balance between network complexity and organisation. The positive correlations suggest that these cities have well-connected and efficient street networks, though less expansive than those in larger cities. These cities often represent historic urban centres with a blend of traditional and modern developments, contributing to their moderately complex networks.

6.1.5 Smaller Urban Areas: Remaining Categories

The remaining clusters represent smaller urban areas across the UK, each with varying degrees of correlation to key topological metrics. For instance, Cluster 1 shows correlations with the proportion of dead ends and 4-way intersections, suggesting a mix of organic and grid-like patterns. Clusters 9 and 10 are notably correlated with self-loop proportion and orientation order, indicating smaller, more contained street networks that emphasise local connectivity. These clusters reflect less complex and less centralised street networks, typical of smaller towns and suburban regions, where local accessibility is prioritised over broader connectivity.

The biserial correlation analysis reveals a clear hierarchy in the complexity and connectivity of street networks across different city categories. The largest cities (Cluster 3) exhibit the most expansive and organised networks, followed by second-tier cities (Cluster 0), and so on. The geographic distribution of these clusters aligns with the hierarchical nature of UK cities, where larger cities function as national and regional hubs, while smaller cities and towns feature more localised networks. Understanding these spatial distributions and topological characteristics is crucial for effective urban planning and development strategies.

Feature Importance via Random Forest We employ a Random Forest model to determine the importance of each topological feature in distinguishing between the clusters. The feature importance plot (Figure 15) ranks the metrics based on their contribution to the model, providing insights into the most influential factors in defining urban categories. Orientation order emerges as the most important feature (importance score: 0.14), highlighting the critical role of street segment alignment and uniformity in distinguishing city categories. The proportion of dead-ends (importance score: 0.12) and clustering coefficient (undirected) (importance score: 0.10) are also highly influential, indicating the significance of street network dead-ends and local clustering in defining city structures. PageRank (importance score: 0.08) and clustering coefficient (directed) (importance score: 0.07) underscore the importance of intersection connectivity and local clustering directionality. Lower importance scores for diameter (0.02) and proportion of 4-way intersections (0.03) suggest these features are less critical in differentiating clusters, indicating they may play more minor roles in the overall topological structure. These insights are summarised in Table 3

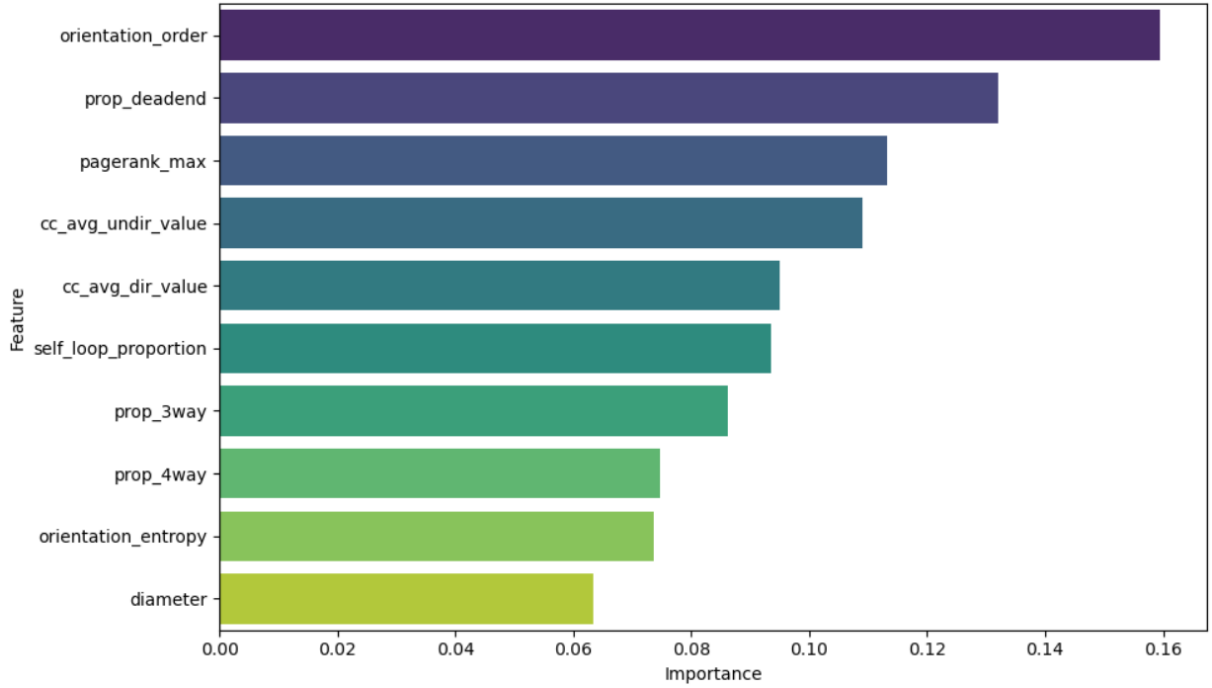


Figure 15: Random Forest - Feature Importance

Feature	Key Insights
Orientation Order	Most important feature (importance score: 0.14), highlighting the significance of alignment and uniformity of street segments.
Proportion of Dead-ends	High importance (0.12), indicating the impact of dead-ends on city structure.
Clustering Coefficient (Undirected)	Significant influence (0.10), reflecting the role of local clustering.
PageRank	Important feature (0.08), underscoring the importance of intersection connectivity.
Clustering Coefficient (Directed)	Important (0.07), indicating the role of directionality in local clustering.
Diameter	Lower importance (0.02), suggesting it is less critical.
Proportion of 4-way Intersections	Lower importance (0.03), suggesting it is less critical.

Table 3: Random Forest - Feature Importance

Principal Component Analysis (PCA) To further assess the key characteristics that distinguish between clusters, we perform Principal Component Analysis (PCA). The PCA explained variance ratio plot (Figure 16) illustrates the cumulative and individual variance explained by each principal component. The first principal component accounts for approximately 37% of the variance, while the first three components together explain around 60% of the variance. This highlights the significance of a few principal components in capturing the most critical aspects of the dataset, particularly those that differentiate the topological features of UK cities. Subsequent components explain progressively less variance, indicating that the earlier components are most effective in distinguishing the unique topological characteristics of each city cluster.

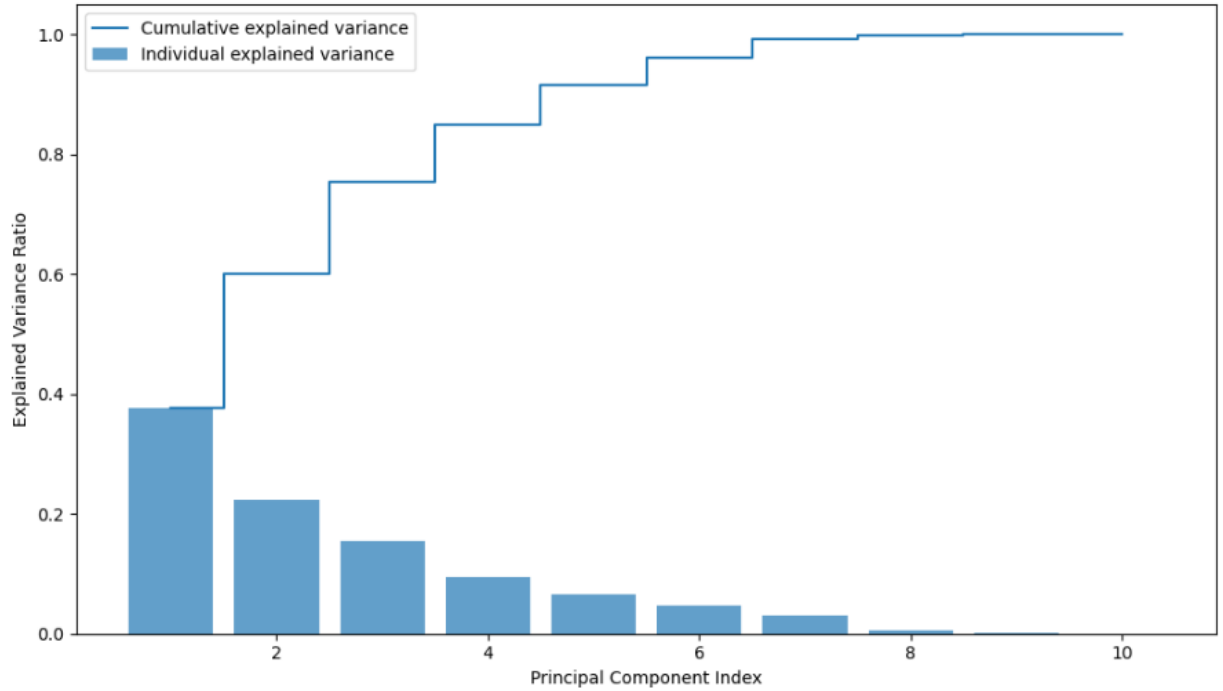


Figure 16: Principal Component Analysis: Explained Variance Ratio

Table 4 presents the *loadings* for each topological metric on the first 10 principal components, highlighting the contributions of these metrics to the overall variance in the dataset. In the context of PCA, *loadings* are the coefficients that indicate the contribution of each original variable to the principal components.

Metric	Comp 1	Comp 2	Comp 3	Comp 4	Comp 5	Comp 6
Diameter (Normalised)	0.32	-0.30	0.025	-0.11	0.62	0.43
Clustering Coefficient	0.25	0.46	0.39	-0.26	0.032	0.073
Proportion 3-Way Intersections	0.40	0.19	-0.19	0.18	-0.089	-0.51
Proportion of 4-Way Intersections	0.33	0.092	-0.47	0.032	-0.25	0.55
Proportion of Dead-Ends	-0.41	-0.21	0.36	-0.11	0.14	-0.0029
PageRank Max	-0.40	0.31	-0.065	0.034	-0.15	0.41
Self-Loop Proportion	0.046	0.17	0.28	0.90	0.21	0.14
Orientation Entropy	-0.16	0.34	-0.43	-0.090	0.68	-0.22
Orientation Order	0.39	-0.40	0.20	-0.019	-0.0054	-0.072

Table 4: Principal Component Analysis Loadings for Topological Metrics (First 6 Components)

The PCA results provide detailed insights into the key dimensions that explain the variance in the topological metrics of UK cities. The first principal component is heavily influenced by diameter (0.32), proportion of 3-way intersections (0.40) and proportion of 4-way intersections (0.33), as well as the clustering coefficients (0.25). This suggests that cities with larger diameters and higher proportions of well-connected intersections tend to exhibit more extensive and interconnected street networks, likely indicative of larger urban areas with complex and integrated street layouts.

The second principal component, is significantly affected by clustering coefficients (0.46), indicating a strong tendency for nodes to form tightly-knit groups. This component also reflects the negative contribution of diameter (-0.30) and proportion of 4-way intersections (-0.47), suggesting that this dimension captures the contrast between densely clustered networks and more widespread urban layouts. The inclusion of orientation entropy (0.34) in PC2 emphasises areas with more randomness in street orientations, possibly indicative of smaller cities or towns with less planned and more organically developed street networks.

Component 3 shows a notable relationship with orientation entropy (-0.43) and proportion of dead-ends (0.36), highlighting a focus on the disorder in street orientations and the prevalence of terminal street segments. This component is inversely related to the clustering coefficients (0.39), suggesting that areas with less orderly street orientations and more dead-ends tend to have lower local clustering, which may characterise suburban or rural areas with less dense and more disorganised street networks.

Component 4 emphasises the self-loop proportion (0.90), suggesting that this component captures variance associated with the occurrence of self-referencing loops within the network, possibly indicative of older city centers with intricate and convoluted street patterns. Additionally, PageRank Max (0.54) is highlighted in PC6, underscoring the centrality and influence of certain nodes within the network, which could indicate areas of high strategic importance within the urban layout.

The PCA results show that a few principal components capture the key aspects of UK cities' topological structures, with the first component explaining approximately 35% of the variance. Metrics like diameter, clustering coefficients and intersection types are crucial in distinguishing city sizes and structures. Larger cities, scoring higher on this component, have more extensive and interconnected networks, indicating well-planned layouts. In contrast, smaller towns and suburbs, with lower scores, exhibit more fragmented and irregular street networks.

These findings align with other analyses, such as Random Forest Feature Importance, which highlight orientation order, proportion of dead-ends and clustering coefficients as key metrics in differentiating urban categories. The consistent focus on street alignment and connectivity across these analyses underscores their importance in defining city structures.

Principal Component	Key Insights
1st Component	Explains approximately 35% of the variance. Highly influenced by diameter, clustering coefficients and proportions of 3-way and 4-way intersections, reflecting extensive and connected street networks.
1st to 3rd Components	Together explain around 60% of the variance. Capture dimensions such as node clustering, orientation entropy and street layout complexity.
Subsequent Components	Explain progressively less variance. Include metrics like self-loop proportion and PageRank Max, indicating the importance of centrality and network loops in older or more intricate urban centers.
Dimensionality Reduction	Reducing the dataset to the first few principal components effectively summarises the data, focusing on the most influential aspects of urban topological structure.

Table 5: Principal Component Analysis: Explained Variance Ratio Table

Category 3 (Largest Cities): These cities are characterised by extensive and highly connected street networks, reflected in high *loadings* on the first principal component. Metrics such as diameter and clustering coefficients suggest dense, well-knit urban structures typical of large metropolitan areas.

Category 0 (Second Largest Cities): These cities maintain high node counts and clustering coefficients, indicating densely connected urban areas, although with slightly less complexity compared to Category 3.

Category 6 (Mid-Sized Cities): Significant positive correlations with orientation order and PageRank Max indicate systematic layouts with strong central areas, typical of mid-sized, strategically important urban hubs.

Category 8 and Category 10 (Smaller Cities): These cities show well-ordered and interconnected networks, with high positive correlations with orientation order and clustering coefficients, suggesting a more organised street layout despite smaller scale.

Category 2 (Smallest Cities): Display high regularity and connectedness in street layout, with high orientation order and clustering coefficients, but with lower node counts and diameters, indicating less extensive networks.

6.2 The Relationship Between Retail Density and Topological Metrics

Regression modeling was employed to explore the relationship between retail density and street network attributes in more detail. Ordinary Least Squares (OLS) regression was initially considered, but given the weak correlations found, more robust methods like Random Forest Regressor and Decision Tree were utilised. These models aimed to capture any potential non-linear relationships and interactions between the metrics.

The results showed that while certain metrics, like average angle and PageRank, had some relevance in specific categories, the overall predictive power of the models was limited. This is evident from the Mean Squared Error (MSE) of 0.00214 and an R-squared value of 0.11. The MSE, which measures the average squared difference between the model's predicted retail density values and the actual values, suggests that the predictions are generally close to the actual values. However, a low R-squared value of 0.11, which indicates how well the model explains the variability in the data, reveals that the model could only account for 11% of the variation in retail density. This suggests that while the errors between predicted and actual values are small on average, the model struggled to capture much of the underlying pattern, limiting its overall accuracy and usefulness.

To further explore the relationship between topological metrics and retail density, a detailed correlation analysis was conducted for each city category. The results were visualised using heatmaps (Figure 17), which showed that no single metric consistently had a strong correlation with retail density across all categories. The green highlights in Figure 17 indicate where correlations were stronger, underscoring the complexity of retail density as a phenomenon likely influenced by a wider range of factors beyond just street network topology.

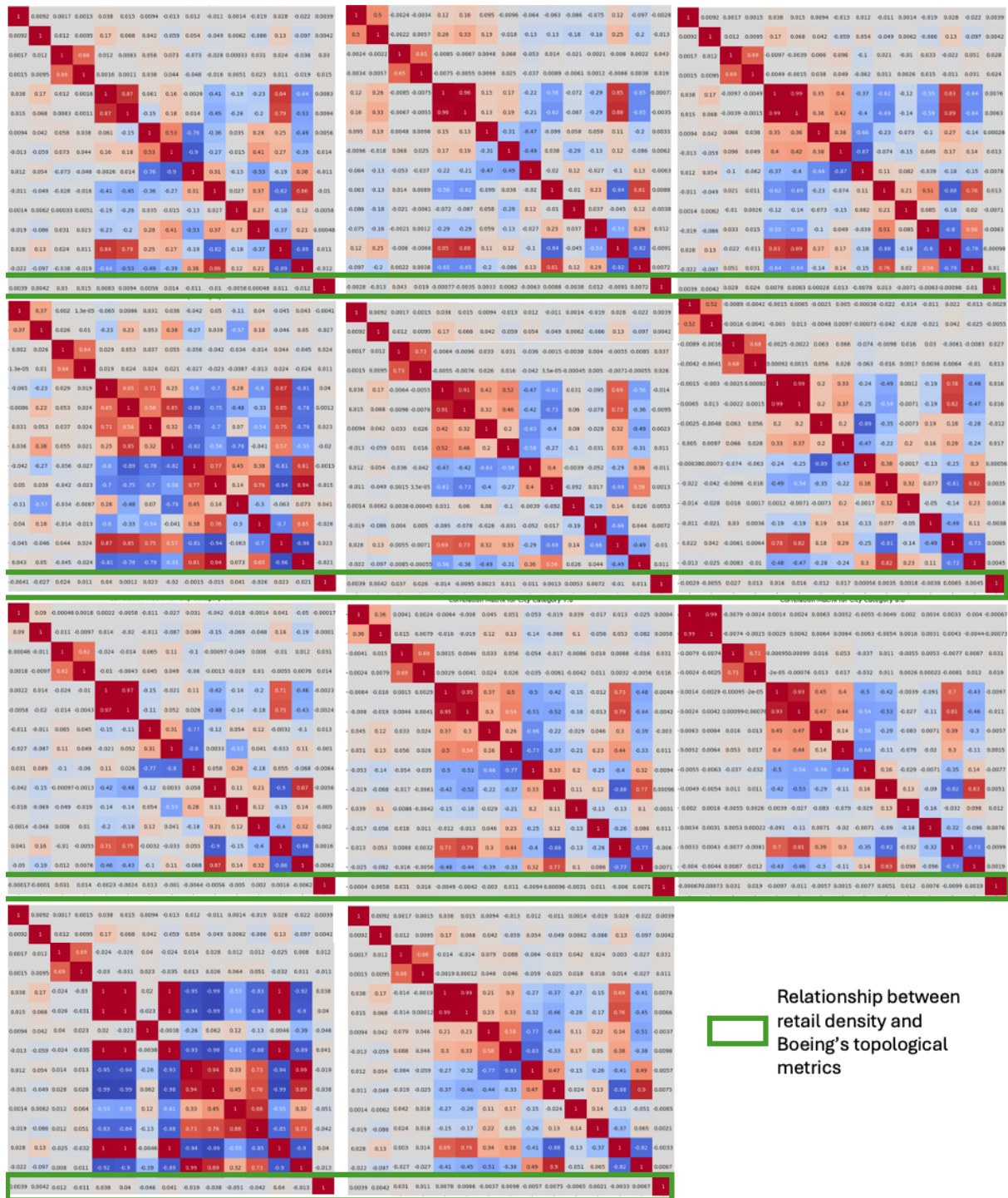


Figure 17: Pearson Correlation Matrices for 11 City Categories Between Topological Metrics and Retail Density. Each matrix represents the correlation between various topological metrics and retail density for a specific city category. The green boxes highlight the correlation values between retail density and topological metrics. Across all city categories, the visualisation demonstrates that there is no significant relationship between topological metrics and retail density, indicating that these metrics do not strongly predict retail density in the analysed cities

6.3 Results

6.3.1 Understanding the Relationship between the topological Network of UK Cities and Retail Density

The primary aim of this study was to assess whether network centrality measures could reliably predict retail density. As discussed earlier, our findings demonstrate that these measures are not effective predictors for the UK street network. To understand why this is the case, we conducted further analysis using spatial autocorrelation and Shannon’s Entropy, which provide a more nuanced explanation of this paradox. Below, we interpret our results in detail.

The model’s limited ability to explain retail density, as indicated by the low R-squared value of 0.11, led us to investigate further why topological metrics alone fall short as predictors. Moran’s I analysis revealed significant spatial autocorrelation in retail locations (Moran’s I: 0.485, p-value: 0.001, z-score: 640.14), indicating that retail sites tend to cluster rather than distribute randomly. This clustering likely reduces the impact of topological metrics on retail density. Moreover, the model’s exclusion of key socio-economic factors, such as population density and income, which are crucial determinants of retail distribution, further limits its explanatory power. Including these variables could offer a more comprehensive understanding of the factors driving retail density in urban areas.

Following the identification of spatial autocorrelation, we calculated Shannon’s Entropy to measure the complexity and unpredictability of street networks in UK cities. The results are visualised in Figure 18, where each point represents a city, with the size of the point indicating edge count and the colour representing the city’s Shannon’s Entropy value. These entropy values, derived from the distribution of street orientations, capture the level of disorder within the street networks, with higher entropy indicating greater unpredictability and complexity. This analysis adds another layer of understanding to the challenges of predicting retail density using topological metrics alone.

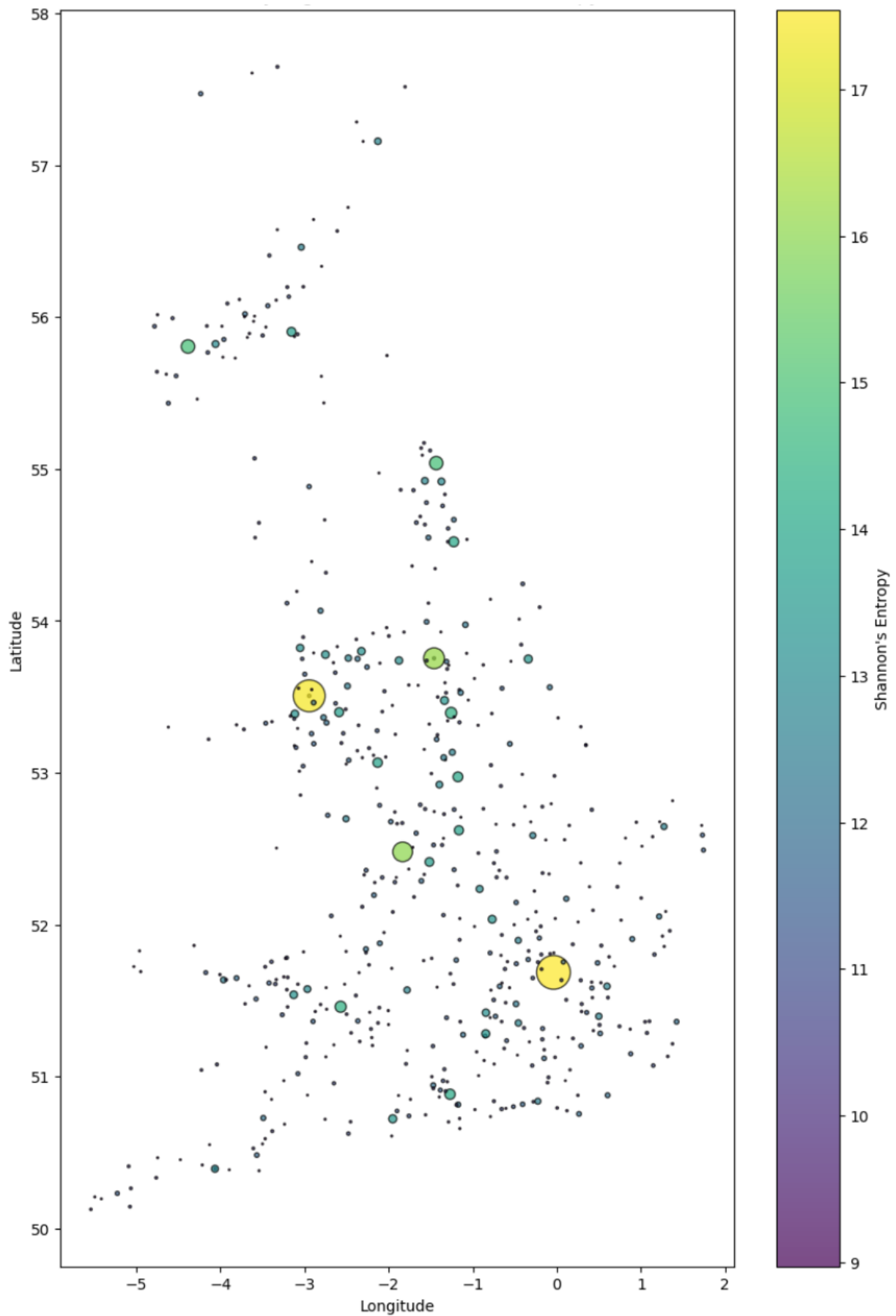


Figure 18: Spatial Distribution of Cities by Size and Shannon's Entropy. This scatter plot visualises cities across the UK, where the size of the points represents city size by edge count and the color gradient indicates Shannon's Entropy. The visualisation highlights the relationship between city size and the complexity of urban spatial structures

The relatively high entropy values observed suggest that the street networks in many city categories are highly irregular and complex. This complexity likely contributes to the difficulty in predicting retail density using traditional centrality metrics.

One of the objectives of this study was to evaluate whether city-level betweenness centrality effectively captures centrality measures. The analysis revealed that while normalising betweenness at the city level provides useful insights into the relative importance of nodes within smaller areas, it may also obscure the broader significance of these nodes within the larger urban network. This normalisation can lead to artificially low or compressed values, reducing the metric’s ability to capture critical connectivity patterns that extend beyond local neighbourhoods. This finding underscores the importance of carefully considering the scale at which centrality measures are applied, as inappropriate normalisation can skew results and diminish the explanatory power of these metrics in understanding urban phenomena like retail density.

7 Discussion

7.1 Clustering as an Explanation of Retail Density

The findings of this dissertation underscore the significant role that spatial autocorrelation plays in predicting retail density within urban environments. While traditional centrality measures, such as betweenness centrality, have been frequently used to forecast retail success, this study suggests that incorporating spatial autocorrelation offers a more nuanced and robust framework for understanding retail distribution, particularly in the context of UK cities. Spatial autocorrelation refers to the degree to which a certain spatial phenomena, such as retail density, is correlated with itself over space. In retail contexts, this implies that the presence of retail establishments in one area can influence the presence and success of retailers in adjacent areas due to factors like pedestrian flow, visibility and accessibility.

This research highlights that centrality measures alone may not fully capture the complexity of retail distribution patterns. While these measures provide valuable insights into potential traffic and accessibility, they do not account for the spatial dependencies that exist between retail locations. Spatial autocorrelation directly addresses these dependencies by analysing how retail density in one area is related to that in nearby areas. This perspective aligns with previous studies that emphasise the importance of spatial autocorrelation in explaining retail clustering and the cumulative effects of customer attraction and competition (Rui et al. 2016).

The practical implications of integrating spatial autocorrelation into urban planning are profound. Recognising the spatial dependencies inherent in retail location decisions allows planners to develop strategies that leverage natural clustering tendencies. For example, policies that encourage the development of retail clusters in areas with existing high spatial autocorrelation can foster more vibrant commercial

districts. Additionally, understanding spatial autocorrelation can help identify potential gaps in retail provision, guiding the strategic placement of new retail establishments in areas where spatial autocorrelation is low but demand is likely to grow (Lan & Liu 2022).

Moreover, this study demonstrates that spatial autocorrelation provides a more comprehensive approach to predicting retail density in the UK than centrality measures alone. Unlike centrality measures, which focus primarily on the network structure of streets, spatial autocorrelation considers the actual distribution of retail activities across the urban landscape. This allows for a more detailed understanding of how retail environments evolve, especially in complex urban settings like those found in the UK. The findings suggest that spatial autocorrelation is a powerful predictor of retail density, offering insights that are both theoretically significant and practically useful for urban planners (Sadahiro 2000).

The integration of spatial autocorrelation with other analytical tools, such as kernel density estimation and network-based methods, further enriches the analysis of retail patterns. For instance, using street segments as the analysis unit, rather than arbitrary grid cells, allows for a more accurate representation of retail distribution along street networks, addressing potential statistical biases. This approach has been shown to effectively reveal the spatial patterns and trends of small business expansion, which are crucial for sustainable urban development (Han et al. 2019a).

7.2 Angular Entropy as an Explanation of Retail Distribution

The concept of angular entropy provides a potential explanation for the observed weak relationship between retail density and betweenness centrality in urban street networks. High angular entropy, which signifies a greater degree of disorder in the street network, may disrupt the expected correlation between centrality measures and retail density Han et al. (2019b), Gudmundsson & Mohajeri (2013). In cities with more complex and less predictable street layouts, the irregularity and unpredictability introduced by high entropy could lead to a more scattered distribution of retail activities, rather than a concentration in areas with high centrality Mohajeri, French & Batty (2013).

This dispersion of retail stores in high-entropy areas might be due to the challenges that irregular street patterns pose for accessibility and navigation, reducing the attractiveness of otherwise central locations for retail development Benoit & Jabari (2021), Pagkalinawan (2019). Consequently, even areas with high betweenness centrality may not experience the expected density of retail activities, as the navigational complexity dissuades both consumers and retailers from clustering in these locations Omer & Kaplan (2019).

In cities with high angular entropy, the expected benefits of centrality may be offset by the increased difficulty in navigating the street network, leading to a less predictable and more dispersed pattern of retail distribution Coutrot et al. (2022).

7.3 Areas for further Research

While this study has highlighted the role of entropy, clustering and spatial autocorrelation in explaining urban retail density in the UK, we have not explored these factors in any detail. Literature suggests this could be due to a variety of factors including socioeconomic data, transport hubs and zoning laws among others. However, we cannot assert this with any certainty in the UK (McConnell et al. 2006, Rothwell & Massey 2008, Tsou & Cheng 2013, Yoshimura et al. 2021). With this in mind further research should investigate diversity of retail establishments, temporal insights, socioeconomic variables and comparisons against other countries. These are explained below.

Temporal Analysis of Orientation Entropy and Retail Dynamics: Future research should focus on the temporal dimension of orientation entropy to understand how changes in street network complexity influence the evolution of retail patterns over time. Longitudinal studies in cities that have undergone significant urban redevelopment or expansion could reveal whether increases in orientation entropy correlate with shifts in retail clustering or dispersion. Such studies would provide valuable insights into the long-term impacts of planned versus organic urban growth on retail vitality, helping to predict how future changes in street networks might shape retail landscapes.

Impact of Orientation Entropy on Different Types of Retail: Further research is needed to investigate how orientation entropy affects various types of retail establishments differently. While this study considers overall retail density, subsequent studies could delve into specific retail sectors, such as high-end boutiques versus convenience stores, to determine whether certain types of retail are more sensitive to the complexity of street networks. This could uncover nuanced dynamics in retail distribution, offering targeted strategies for urban planners to support different retail sectors based on their relationship with street network entropy.

Integration of Socio-Economic Variables with Entropy Measures: To develop a more comprehensive model of retail distribution, future research could attempt to integrate socio-economic variables such as income levels, pedestrian traffic and property values with orientation entropy. This would help to identify whether the impact of street network complexity on retail density is moderated by these socio-economic factors, providing a richer understanding of the socio-spatial dynamics in urban environments. Such an approach could lead to more effective urban planning policies that consider both the physical structure of cities and the social and economic context in which retail operates.

Comparative Studies Across Different National Contexts: Given the UK-specific focus of this study, future research applying our research techniques to replicate our analysis in cities in different countries with varying urban planning histories and street network structures would be valuable to validate our approach.. By comparing how orientation entropy influences retail distribution across diverse national contexts, researchers could determine whether the relationships observed in UK cities are universally applicable or context-specific. This would contribute to a broader understanding of the role of

street network complexity in shaping retail patterns on a global scale, offering insights that could inform international urban planning practices.

7.4 Concluding Thoughts

In conclusion, this dissertation emphasises the critical importance of spatial autocorrelation in understanding and predicting retail density. By incorporating spatial autocorrelation into urban analysis, researchers and planners can gain a deeper insight into the factors that drive retail success, leading to more effective and contextually appropriate planning strategies. As urban science continues to evolve, the integration of spatial autocorrelation with other spatial analysis techniques will be essential for developing comprehensive models to accurately predict retail density in diverse urban settings. This holistic approach will ultimately contribute to the creation of more resilient, dynamic and equitable urban retail landscapes.

A Appendix: Research Log

Date	Meeting Overview
12 Apr 2024	Initial presentation of research question and methods to be used for research.
22 Apr 2024	Academic Supervisor (Dr. Elsa Arcaute) and Industry Advisor (Mr. Ed Parham) meet to discuss Space Syntax Ltd. and UCL Coordination for the duration of the research process.
2 May 2024	First meeting with Academic Advisor at the Alan Turing Institute. Discussed general direction, specifics of the clustering methods and percolation process that will be used in analysis. Supervisor advised several research papers which would be used including. Academics included Elsa Arcaute, Valentina Marin, Carlos Molinero, Carlo Ratti, Geoff Boeing, Sergio Porta and Roberto Murcio
17 May 2024	Presented initial results on our research to UCL Centre for Advanced Spatial Analysis doctoral students and to receive feedback and guidance on future direction of
14 June 2024	Had conversation with with Dr. Gareth Simons. Dr. Simons advised the used of hierarchical clustering to categorise cities to highlight the implicit hierarchies in UK Cities. In addition, he advised using OpenRoads as the main dataset for analysis due to its complexity and detail. Discussed PCA and Cityseer to assign POIs.
1 July 2024	Presented second round of findings and first analysis and results of the topic. This included Occupation Probability, Time-Based Percolation, Distance-Based Percolation and Definition of the Angular Network all at different scales. Further discussed the nuances between of distance, angular and time based percolatio and how each of these relates to urban development and economic activity. Discussed the role of diameter, node count and clustering in retail density and urban categorisation.
25 July 2024	Check-in with Academic Supervisor on progress to date. Determined Boeing metrics as the metrics to be used in the assessment of urban areas and to assess clusters.

7 Aug 2024	Meeting with UCL Centre for Advanced Spatial Analysis doctoral student Andrew Renninger to discuss entropy and spatial autocorrelation as a possible explanation for the weak relationship between betweenness centrality and retail density.
10 Aug 2024	Finalised narrative to be written in the paper - concluded on final research questions and suggestions on visualisations.

B Appendix: Scripts

```
# Function to convert NetworkX graph to graph-tool graph

def nx_to_gt(G_nx):
    G_gt = gt.Graph(directed=False)
    vertex_map = {}

    pos = G_gt.new_vertex_property("vector<double>")
    original_ids = G_gt.new_vertex_property("string")
    angles = G_gt.new_vertex_property("double")
    form_of_road_node = G_gt.new_vertex_property("string")
    geometry = G_gt.new_vertex_property("string")
    degree = G_gt.new_vertex_property("int")

    for node, data in tqdm(G_nx.nodes(data=True), desc="Converting nodes
        from NetworkX to graph-tool"):
        v = G_gt.add_vertex()
        vertex_map[node] = v
        pos[v] = list(map(float, data['pos'].strip('()').split(',')))
        original_ids[v] = node
        angles[v] = data.get('angle', 0) # Default to 0 if angle not
            present
        form_of_road_node[v] = data.get('form_of_road_node', '')
        geometry[v] = data.get('geometry', '')
        degree[v] = data.get('degree', 0)

    length = G_gt.new_edge_property("double")
    original_edge_ids = G_gt.new_edge_property("object")
    road_classification = G_gt.new_edge_property("string")
    road_function = G_gt.new_edge_property("string")
    form_of_way = G_gt.new_edge_property("string")
    primary_route = G_gt.new_edge_property("bool")
    trunk_road = G_gt.new_edge_property("bool")
    edge_geometry = G_gt.new_edge_property("string")
```

```

for u, v, data in tqdm(G_nx.edges(data=True), desc="Converting edges
    from NetworkX to graph-tool"):
    e = G_gt.add_edge(vertex_map[u], vertex_map[v])
    length[e] = data.get('length', 0.0)
    original_edge_ids[e] = (u, v)
    road_classification[e] = data.get('road_classification', '')
    road_function[e] = data.get('road_function', '')
    form_of_way[e] = data.get('form_of_way', '')
    primary_route[e] = data.get('primary_route', False)
    trunk_road[e] = data.get('trunk_road', False)
    edge_geometry[e] = data.get('geometry', '')

G_gt.vertex_properties['pos'] = pos
G_gt.vertex_properties['original_id'] = original_ids
G_gt.vertex_properties['angle'] = angles
G_gt.vertex_properties['form_of_road_node'] = form_of_road_node
G_gt.vertex_properties['geometry'] = geometry
G_gt.vertex_properties['degree'] = degree

G_gt.edge_properties['length'] = length
G_gt.edge_properties['original_edge_ids'] = original_edge_ids
G_gt.edge_properties['road_classification'] = road_classification
G_gt.edge_properties['road_function'] = road_function
G_gt.edge_properties['form_of_way'] = form_of_way
G_gt.edge_properties['primary_route'] = primary_route
G_gt.edge_properties['trunk_road'] = trunk_road
G_gt.edge_properties['geometry'] = edge_geometry

return G_gt, vertex_map

# Convert your NetworkX graph G_urbclus to a graph-tool graph
G_gt, vertex_map = nx_to_gt(G_urbclus)

```

Listing 1: Convert a NetworkX Graph into a *Graph_{tool}Graph*

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